

Foc_2024b Detailed Design Document (AI-Enhanced)

**Source workbook: /Users/wangtianqi/
Desktop/autoModeling/work/tst_mdI/
Foc_2024b_signals_cal.xlsx Generated:
2026-06-18 23:32:20 Analysis depth: 86
subsystems**

wangtianqi

18-6月-2026

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1. Subsystem Navigation

Click a subsystem name to jump to its corresponding report section.

Subsystem	Role	Block Types
Foc_2024b		
FieldOrientedControl		
ClarkTransformations		
CurrentControl		
MOTOR_STATE_ALIGNMENT		
CompareToConstant		
EnableDriver		
SetPiController		
PiCtrl_Calc_D		
PiControl		
PiCtrl_Calc_q		
MOTOR_STATE_BACK		
PiCtrl_Calc_Q		
parkangle_cal		
ParkAngleCcw		
ParkAngleCw		
MOTOR_STATE_BRAKE		
MOTOR_STATE_CLOSED_LOOP		
PiCtrl_Calc		
CalcVqMax		
GetVqMax		
VaMax_ValidIndex		
Q_Mul_s16s16_s16		
VqMax_InvalidIndex		

1. Subsystem Navigation

Index		
SaturationDynamic		
VqMaxTable		
PIControl_update		
PI_update		
MOTOR_STATE_RAMP		
MOTOR_STATE_RAMP_INIT		
default		
CurrentOffsets		
OffsetDefault		
OffsetDetection		
BitShift		
bit_shift		
BitShift1		
GetCurrents		
AdcVolToCur		
DEFAULT		
SECTOR_1		
SECTOR_2		
SECTOR_3		
SECTOR_4		
SECTOR_5		
SECTOR_6		
InverseClarkTransformations		
InverseParkTransformations		
ParkTransformations		
PhaseCurrentSpikeMonitor		
b_fault_debounce		

1. Subsystem Navigation

PIIObserver		
BackEMF		
AlphaBackEMF		
BetaBackEMF		
LowPassFilter		
ParkTransformationBEMF		
SinCosCalculation		
SupplyVoltageAdaptive		
ThetaOmegaCalculation		
SendCurrentToCan		
Subsystem		
Subsystem1		
SetDutyCycle		
PreCondition		
FINAL_OFF		
HwModeActive		
HwModeInactive		
STATE_OFF		
SectorDetectionPart1		
Sector_3		
Sector_5_1		
Sector1		
Sector5		
SectorDetectionPart2		
Sector_4		
Sector_6_2		
Sector2		
Sector6		

1. Subsystem Navigation

SetDuty		
Default		
CalcPwmShift		
Sector3		
Sector4		
foc_calc_sin_table		

2. Model Overview

【Foc_2024b 模型概述】

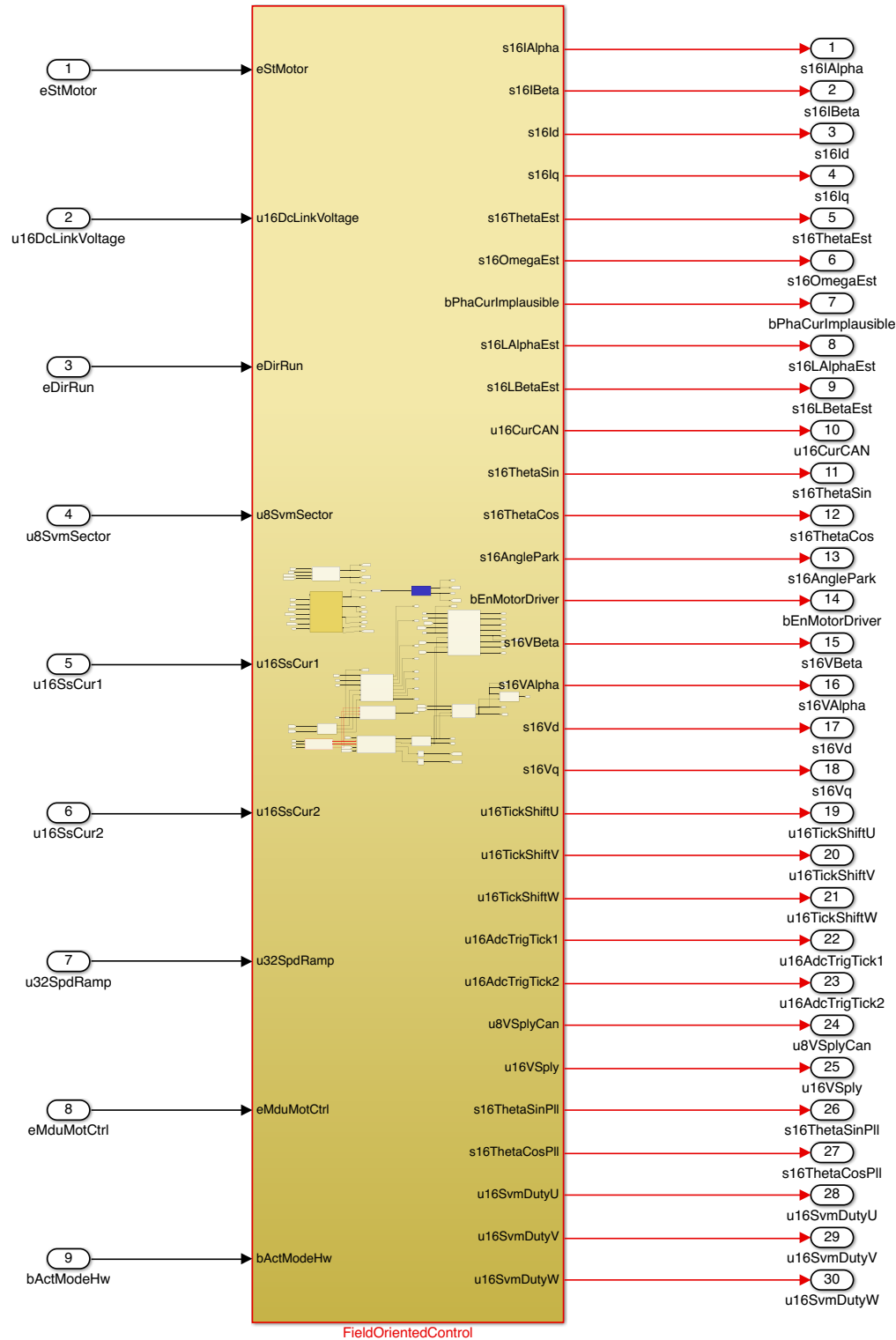
本模型共包含 127 个子系统模块，已深度分析 86 个。

以下是各子系统的功能分布：

- Foc_2024b:
- FieldOrientedControl:
- ClarkTransformations:
- CurrentControl:
- MOTOR_STATE_ALIGNMENT:
- CompareToConstant:
- EnableDriver:
- SetPiController:
- PiCtrl_Calc_D:
- PiControl:
- PiCtrl_Calc_q:
- MOTOR_STATE_BACK:
- PiCtrl_Calc_Q:
- parkangle_cal:
- ParkAngleCcw:

...及其他 71 个子系统（详情见各子系统章节）

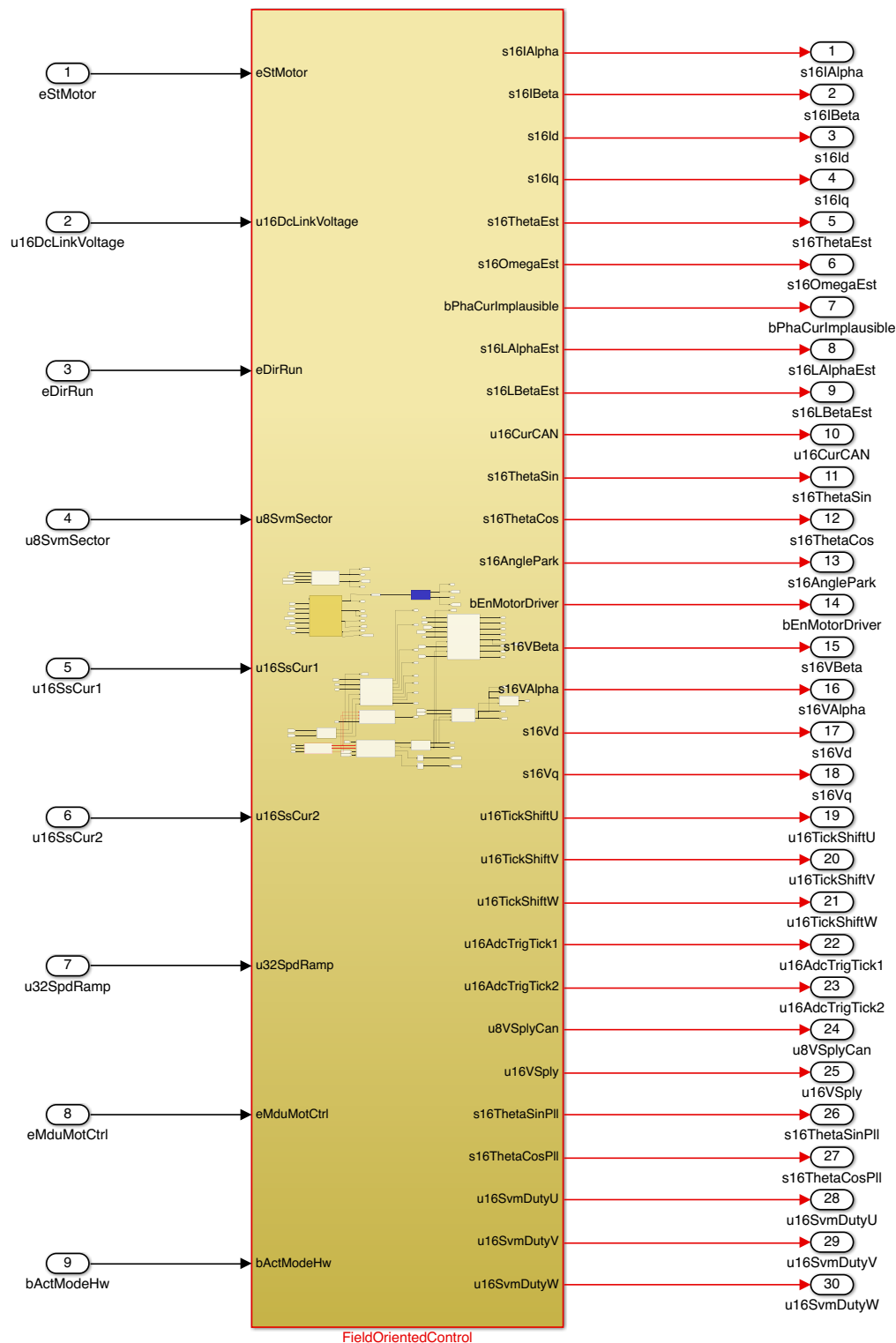
2. Model Overview



3. Foc_2024b

【Foc_2024b】信号处理与功能传递模块。

3. Foc_2024b

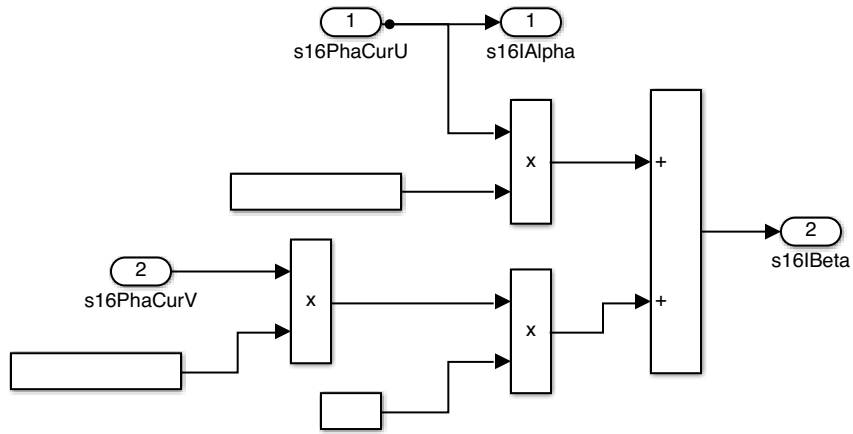


【FieldOrientedControl】 信号处理与功能传递模块。



5. ClarkTransformations

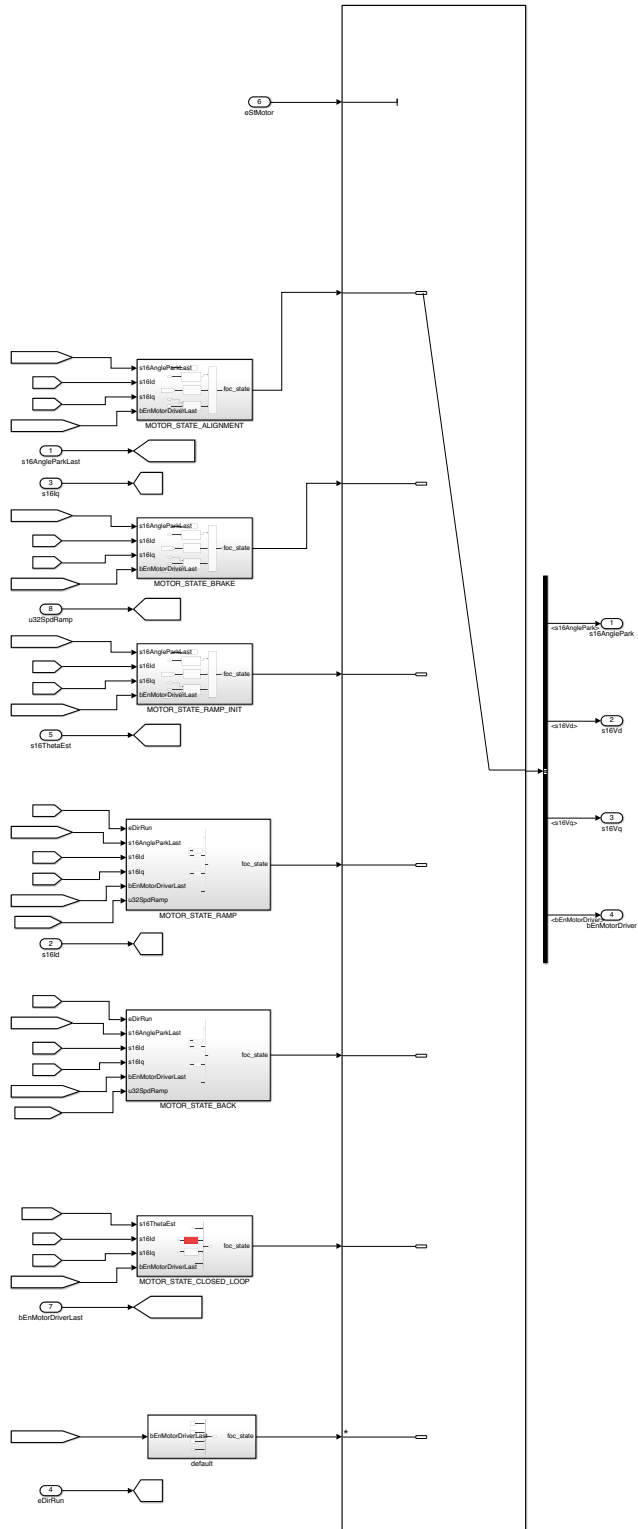
【ClarkTransformations】 信号处理与功能传递模块。



6. CurrentControl

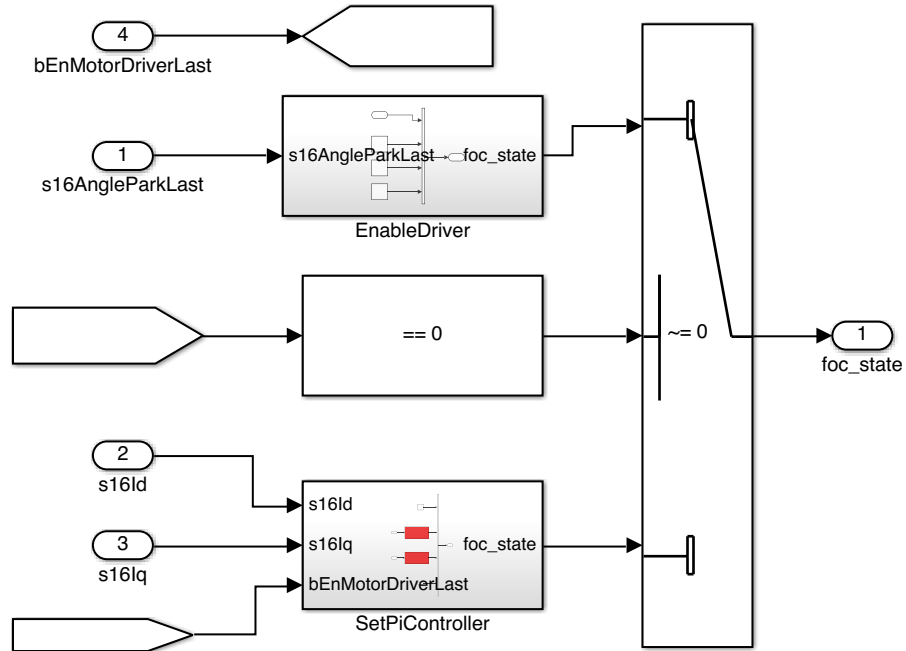
【CurrentControl】信号处理与功能传递模块。

6. CurrentControl



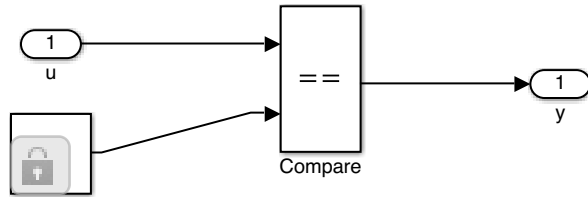
7. MOTOR_STATE_ALIGNMENT

【MOTOR_STATE_ALIGNMENT】信号处理与功能传递模块。



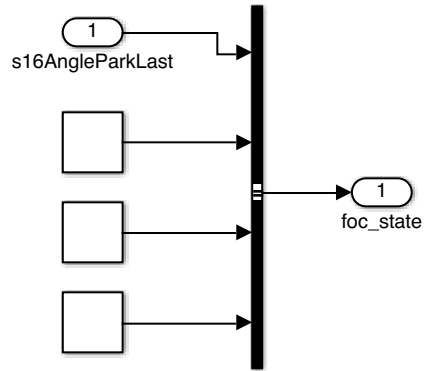
8. CompareToConstant

【CompareToConstant】信号处理与功能传递模块。



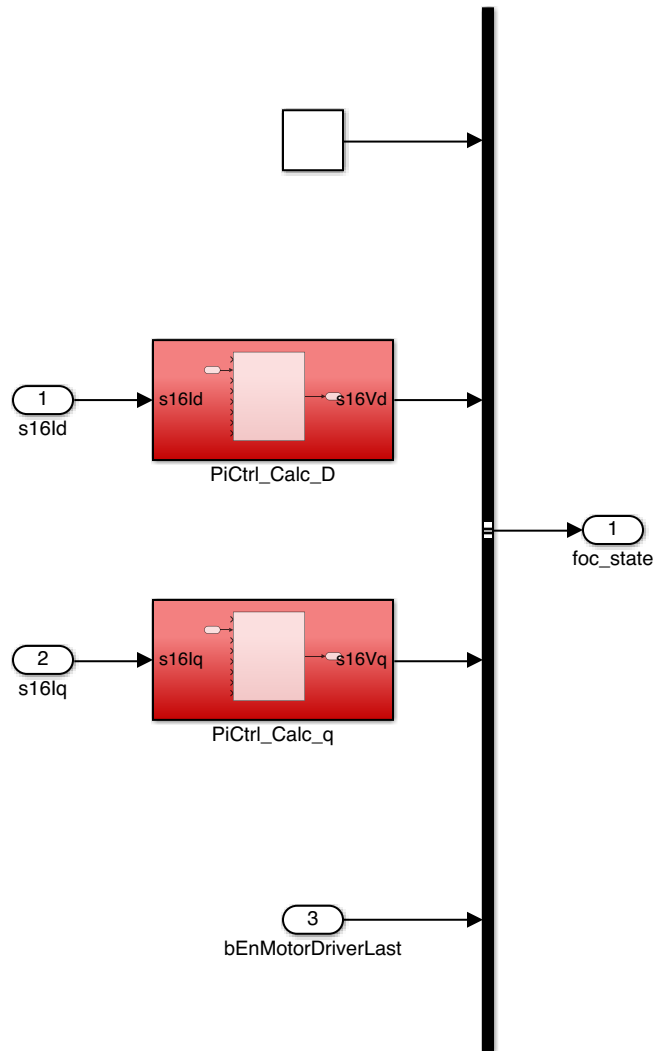
9. EnableDriver

【EnableDriver】信号处理与功能传递模块。



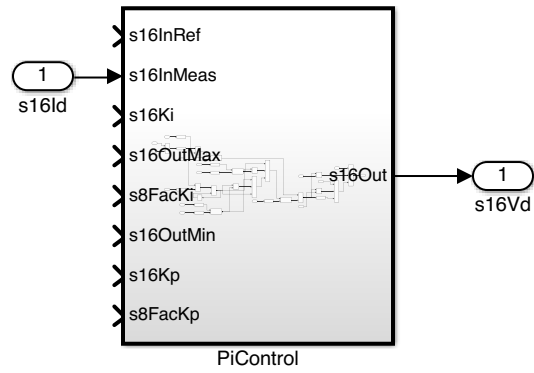
10. SetPiController

【SetPiController】信号处理与功能传递模块。



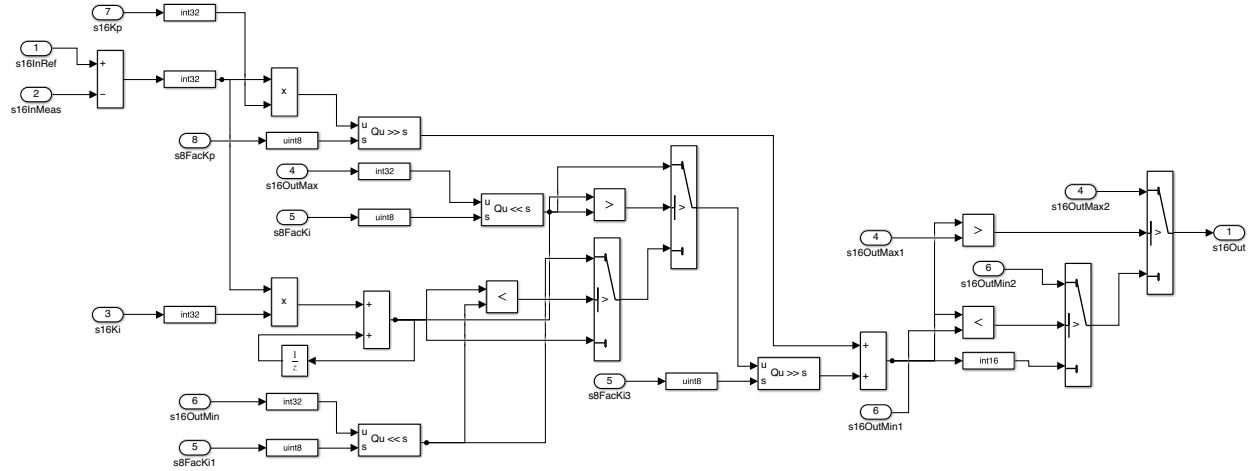
11. PiCtrl_Calc_D

【PiCtrl_Calc_D】信号处理与功能传递模块。



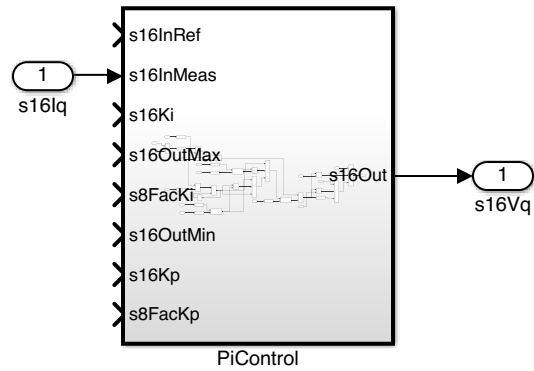
12. PiControl

【PiControl】信号处理与功能传递模块。



13. PiCtrl_Calc_q

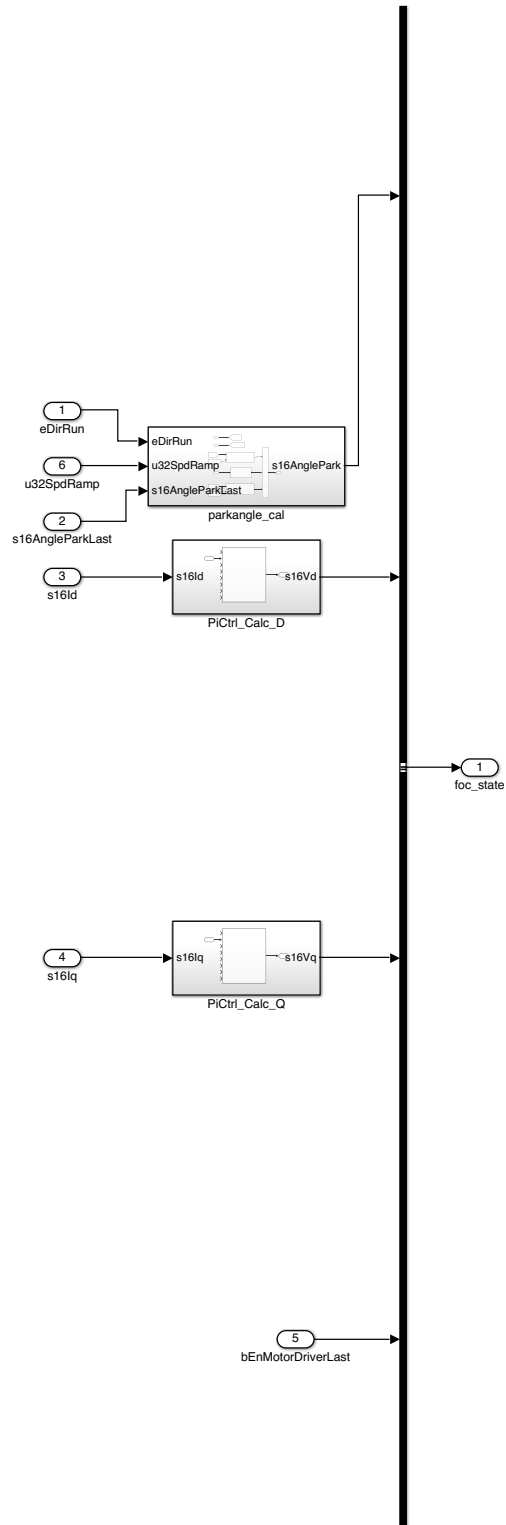
【PiCtrl_Calc_q】信号处理与功能传递模块。



14. MOTOR_STATE_BACK

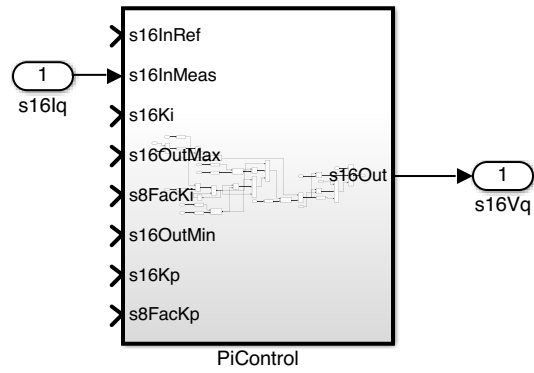
【MOTOR_STATE_BACK】信号处理与功能传递模块。

14. MOTOR_STATE_BACK



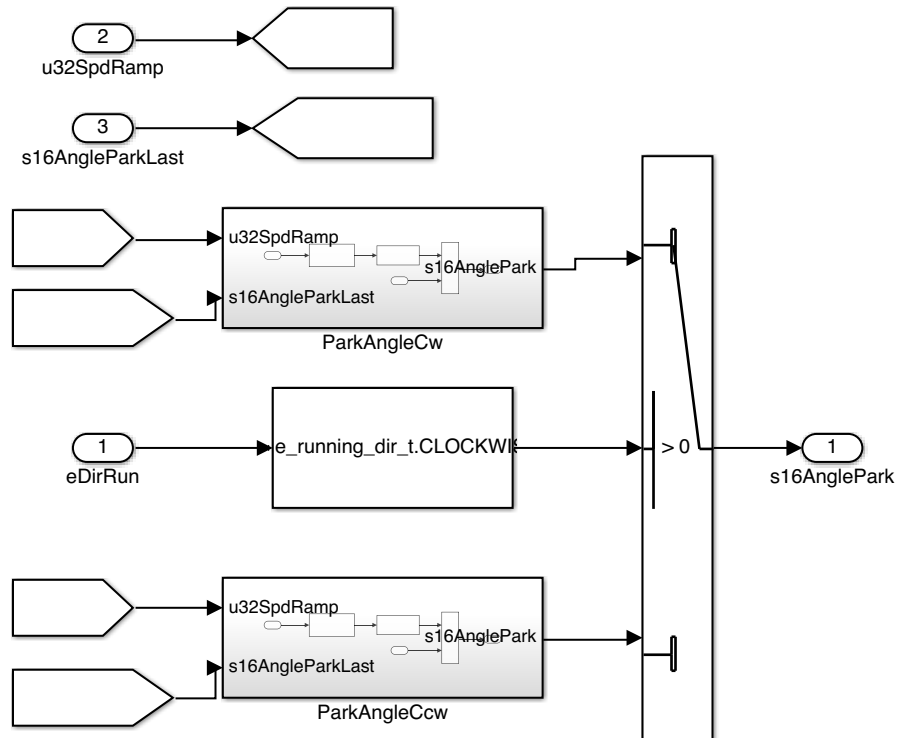
15. PiCtrl_Calc_Q

【PiCtrl_Calc_Q】信号处理与功能传递模块。



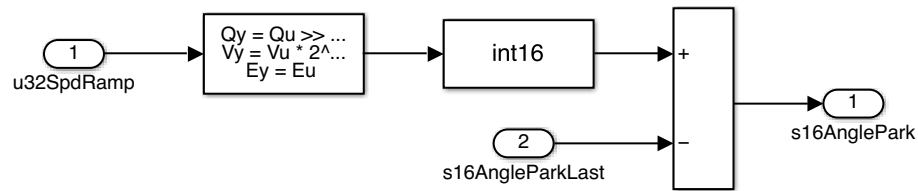
16. parkangle_cal

【parkangle_cal】信号处理与功能传递模块。



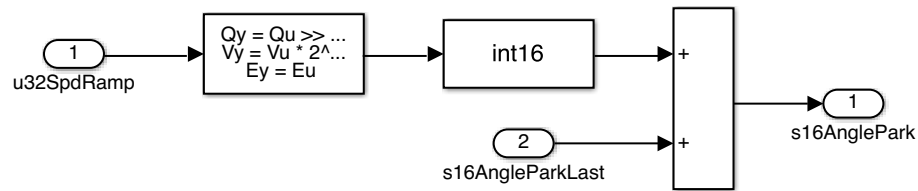
17. ParkAngleCcw

【ParkAngleCcw】 信号处理与功能传递模块。



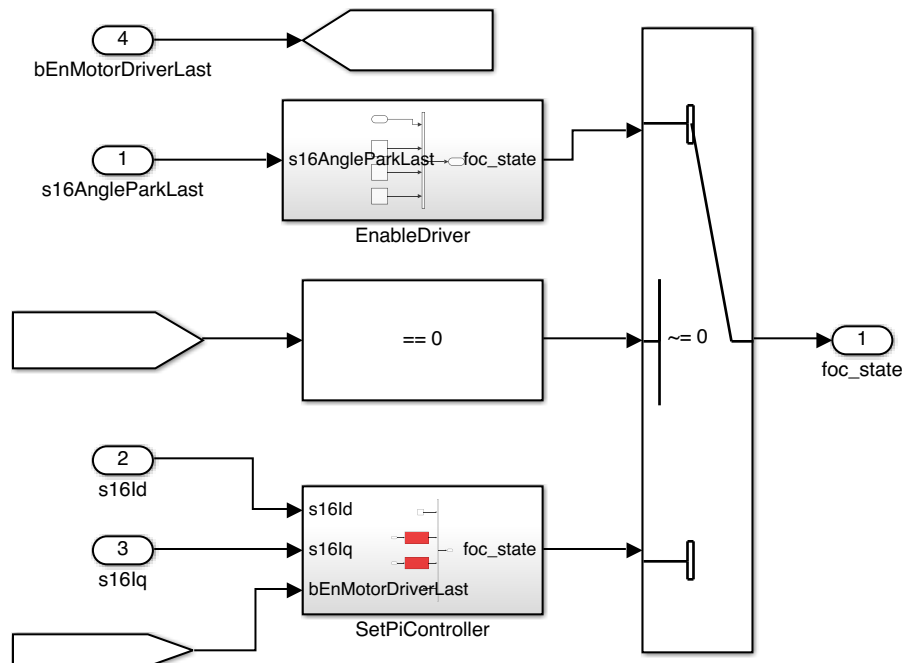
18. ParkAngleCw

【ParkAngleCw】 信号处理与功能传递模块。



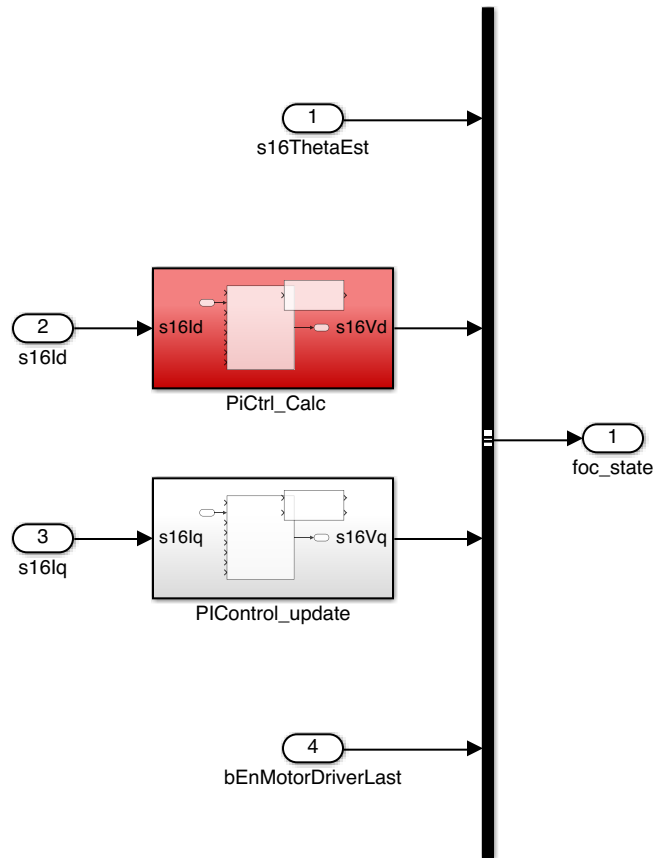
19. MOTOR_STATE_BRAKE

【MOTOR_STATE_BRAKE】信号处理与功能传递模块。



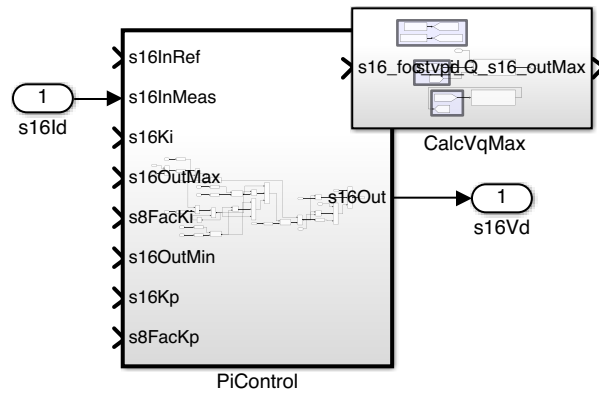
20. MOTOR_STATE_CLOSED_LOOP

【MOTOR_STATE_CLOSED_LOOP】信号处理与功能传递模块。



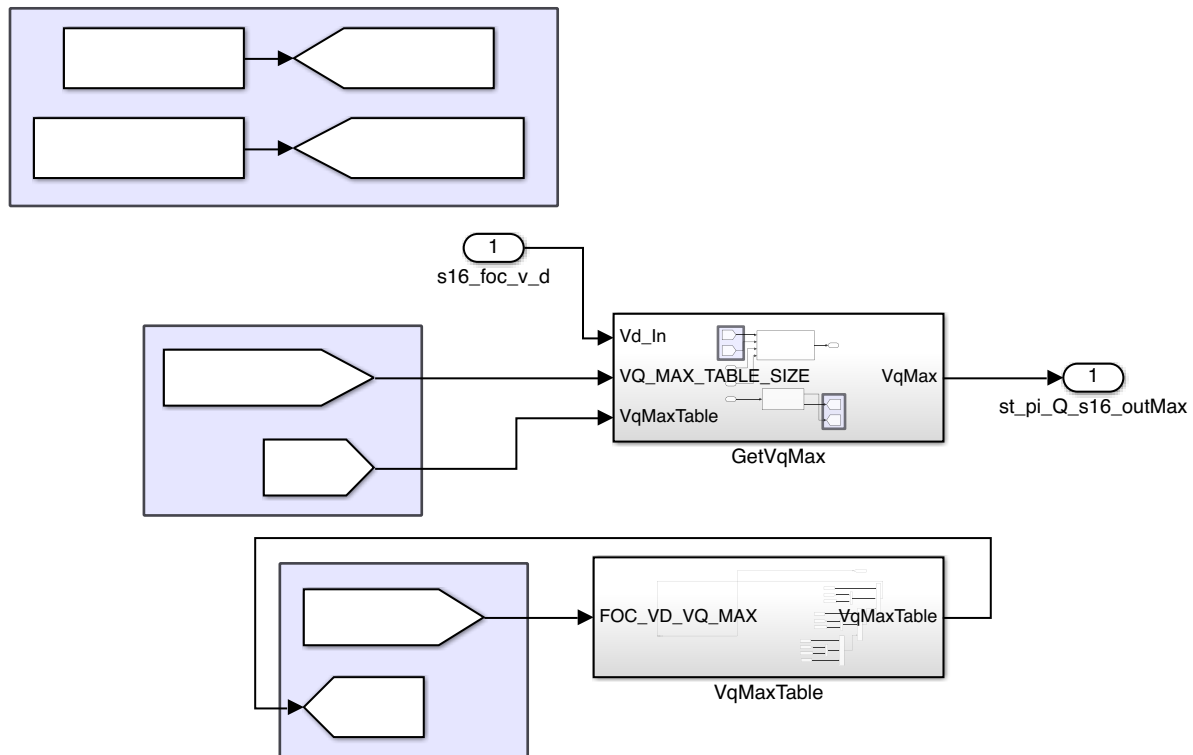
21. PiCtrl_Calc

【PiCtrl_Calc】信号处理与功能传递模块。



22. CalcVqMax

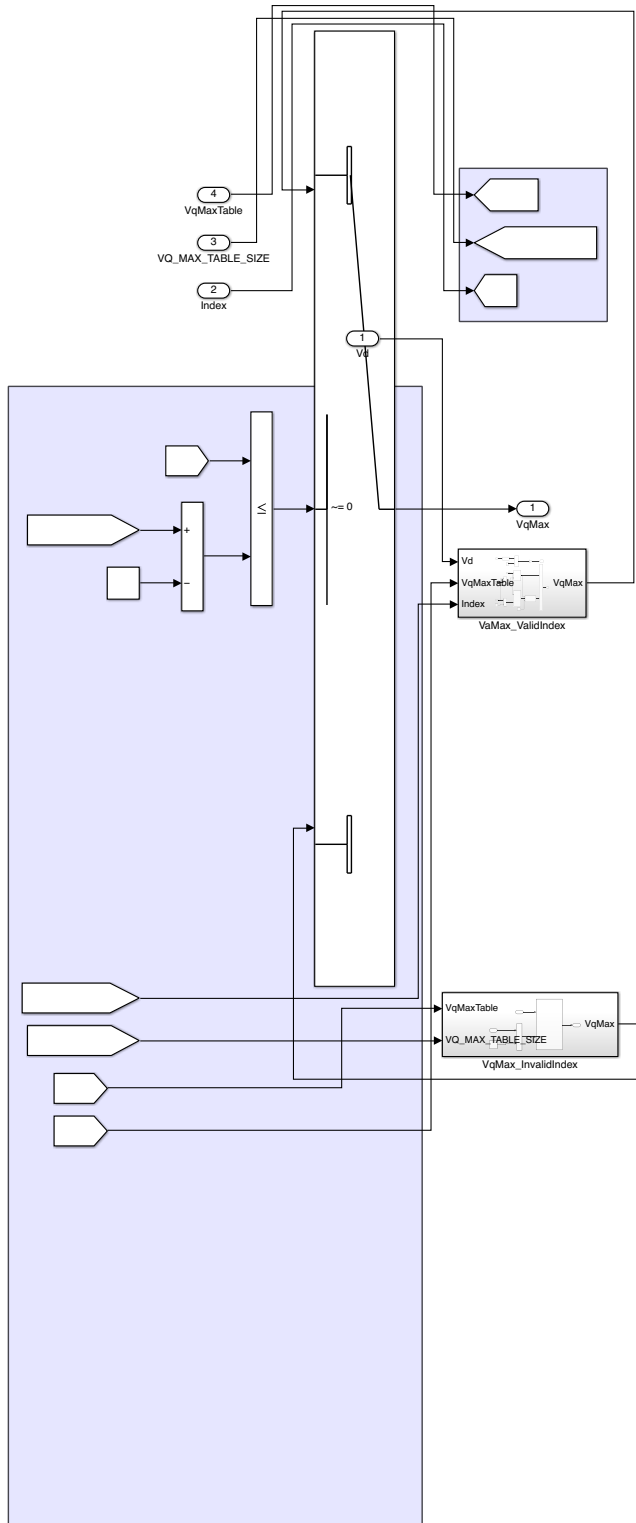
【CalcVqMax】信号处理与功能传递模块。



23. GetVqMax

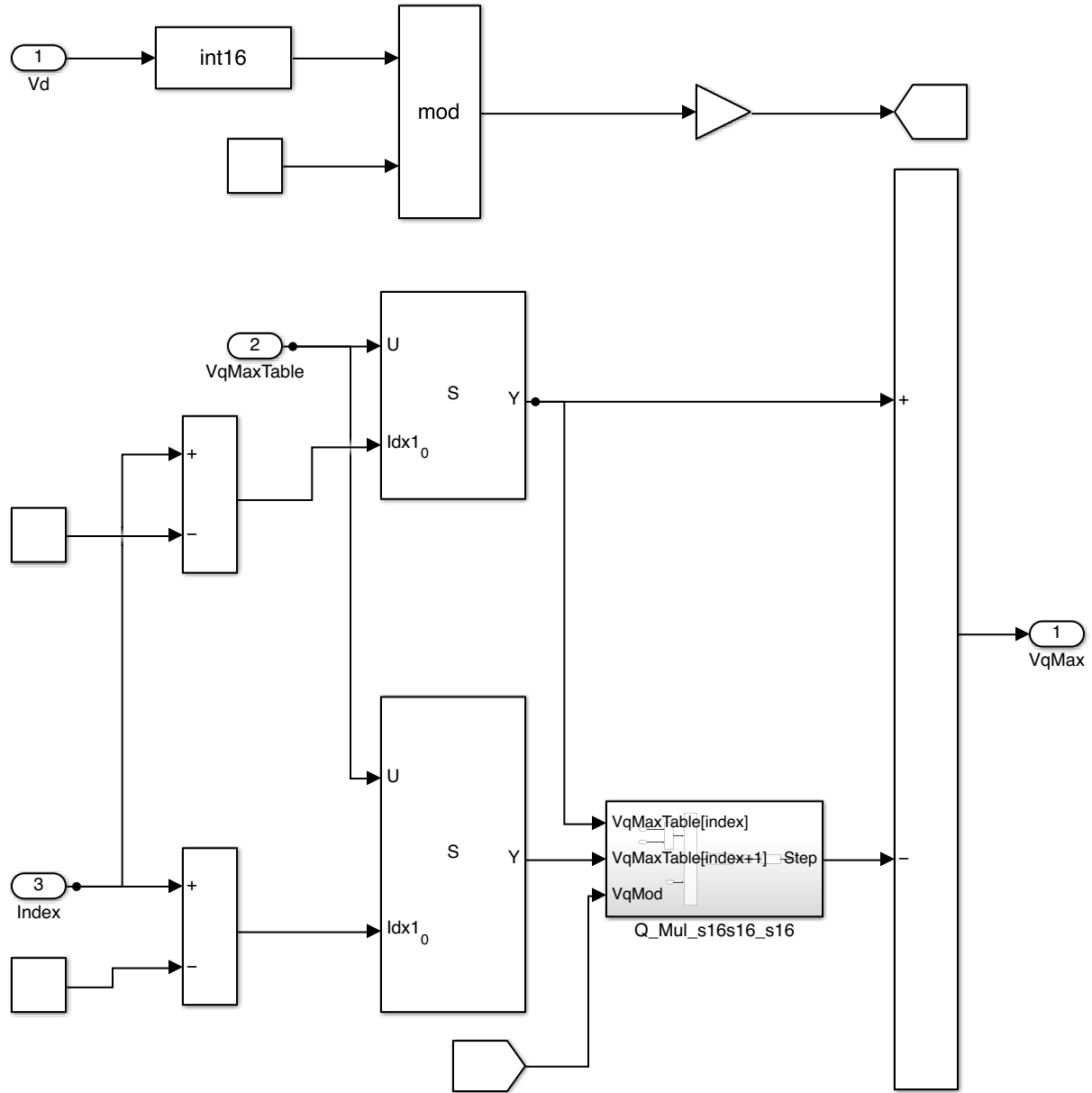
【GetVqMax】信号处理与功能传递模块。

23. GetVqMax



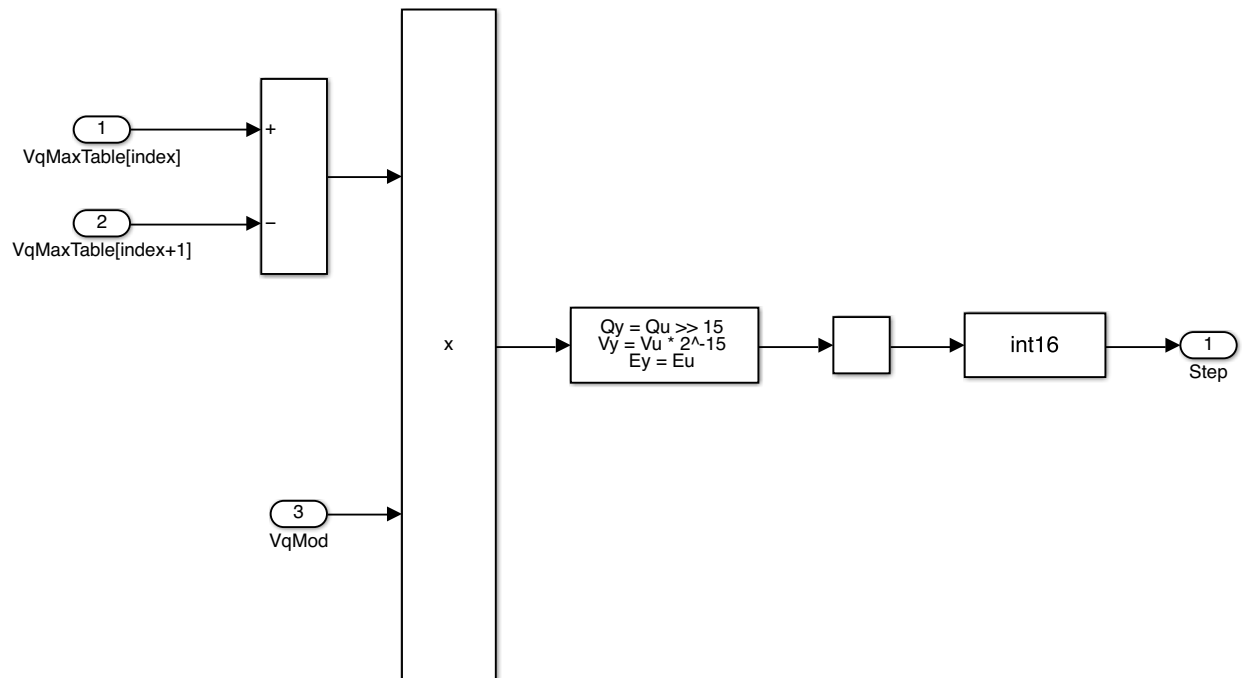
24. VaMax_ValidIndex

【VaMax_ValidIndex】信号处理与功能传递模块。



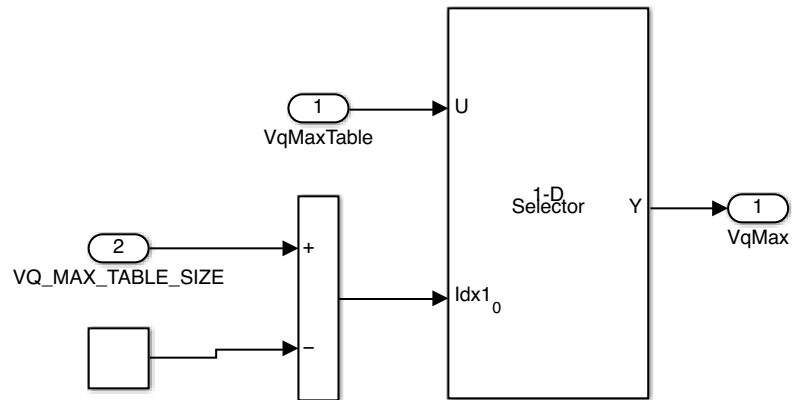
25. Q_Mul_s16s16_s16

【Q_Mul_s16s16_s16】信号处理与功能传递模块。



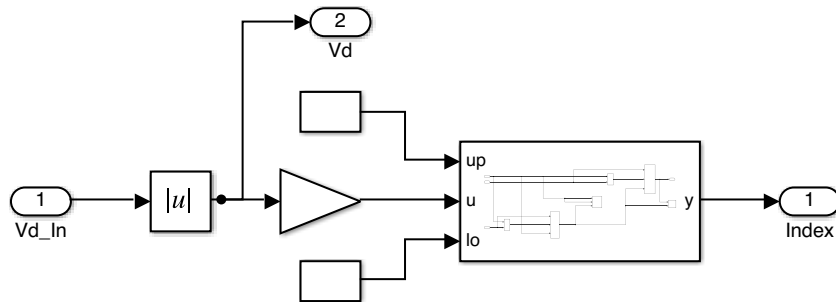
26. VqMax_InvalidIndex

【VqMax_InvalidIndex】信号处理与功能传递模块。



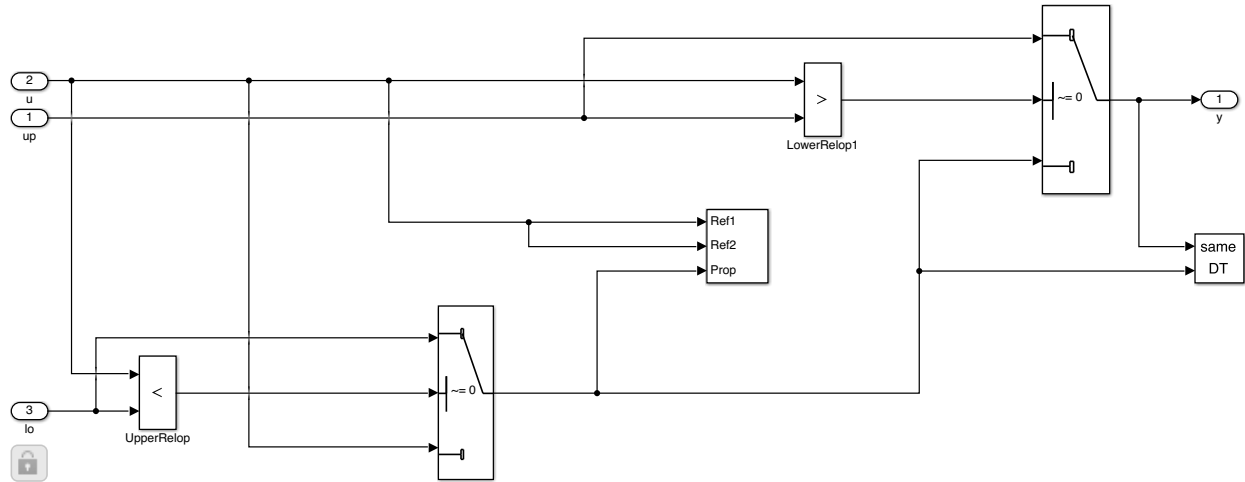
27. Index

【Index】 信号处理与功能传递模块。



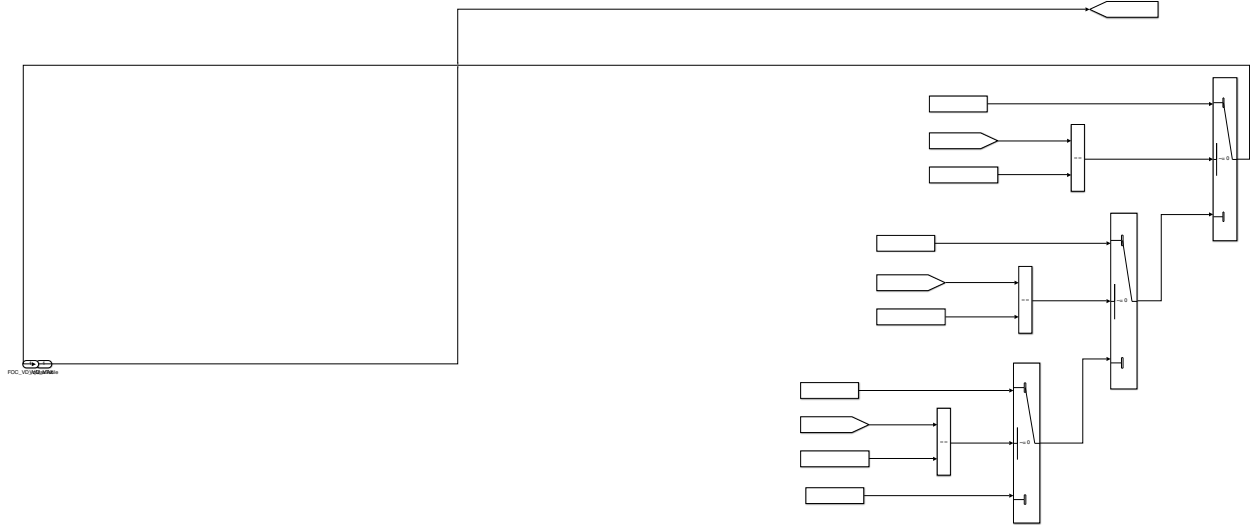
28. SaturationDynamic

【SaturationDynamic】 信号处理与功能传递模块。



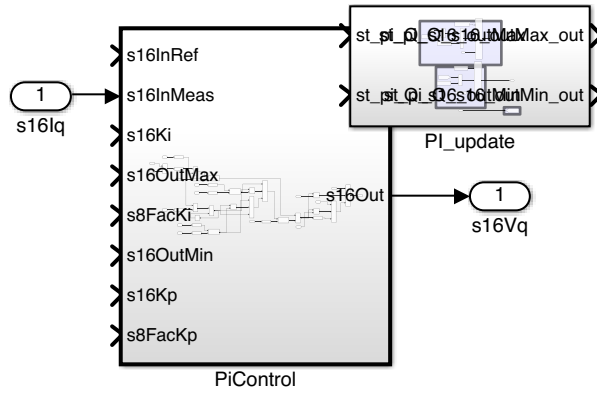
29. VqMaxTable

【VqMaxTable】信号处理与功能传递模块。



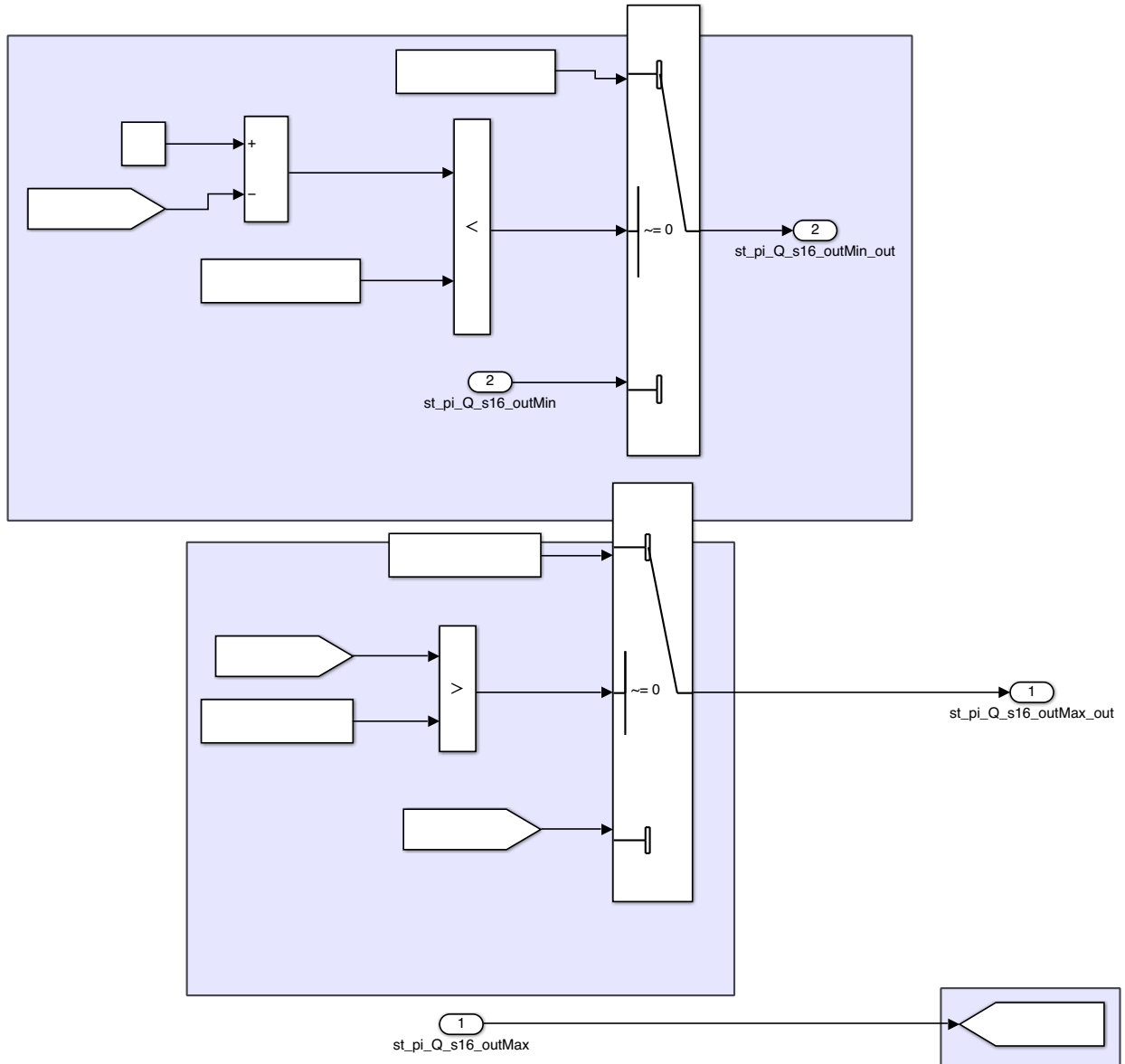
30. PIControl_update

【PIControl_update】信号处理与功能传递模块。



31. PI_update

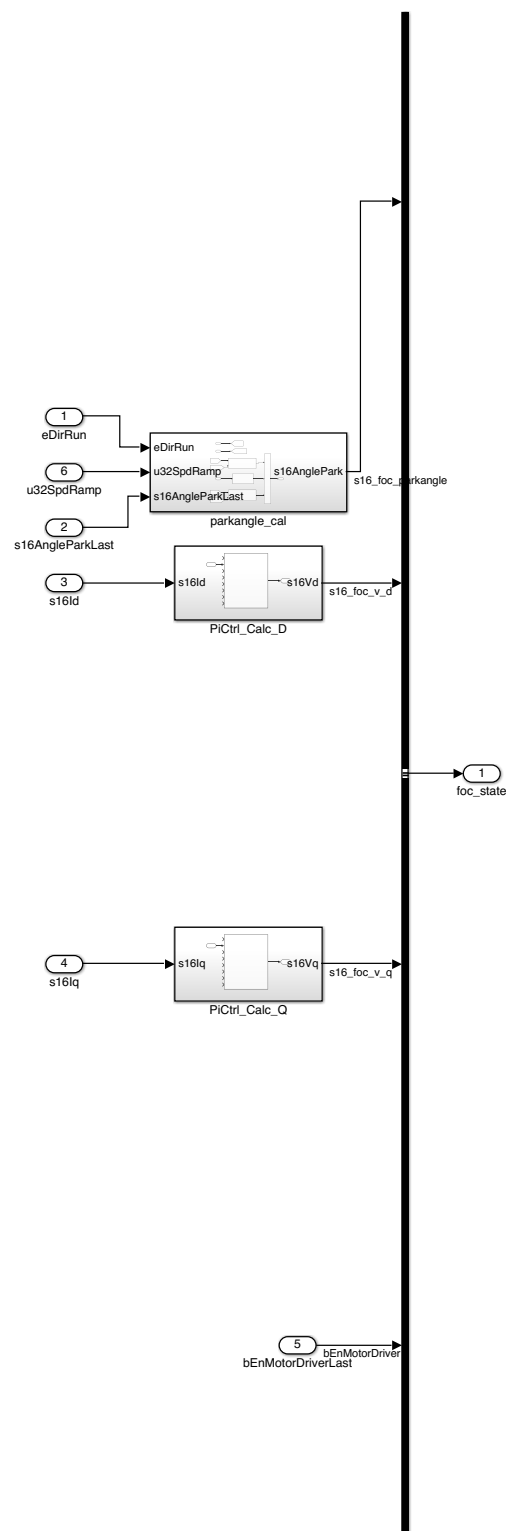
【PI_update】信号处理与功能传递模块。



32. MOTOR_STATE_RAMP

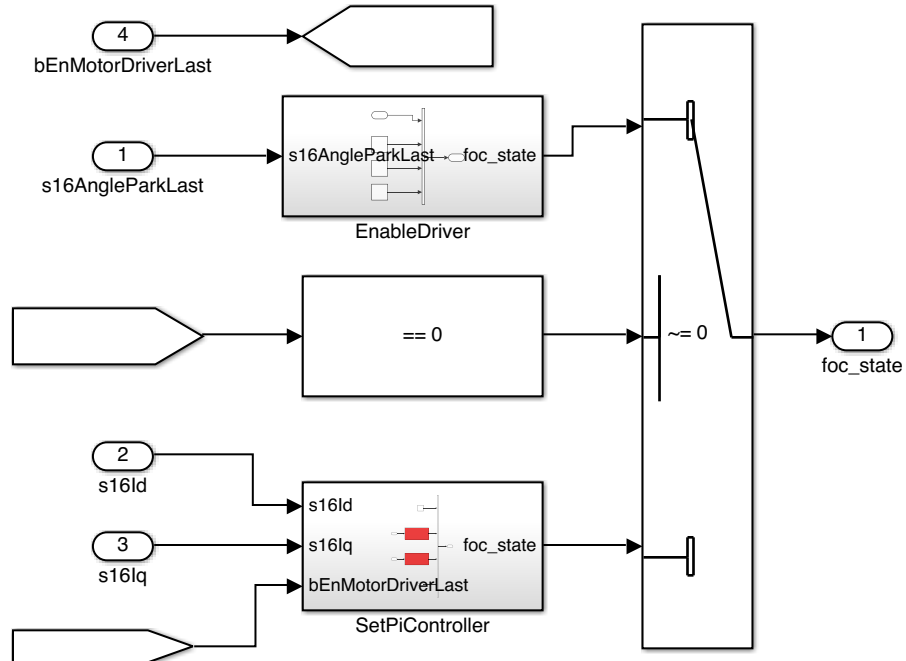
【MOTOR_STATE_RAMP】 信号处理与功能传递模块。

32. MOTOR_STATE_RAMP



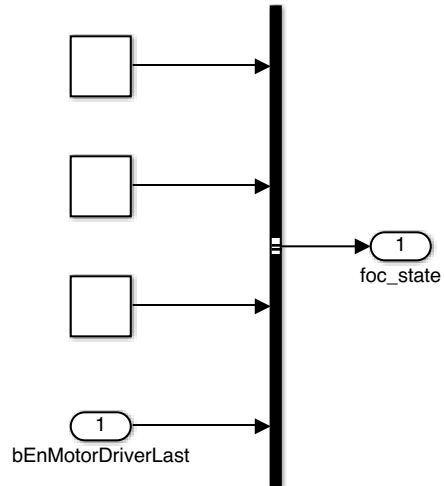
33. MOTOR_STATE_RAMP_INIT

【MOTOR_STATE_RAMP_INIT】信号处理与功能传递模块。



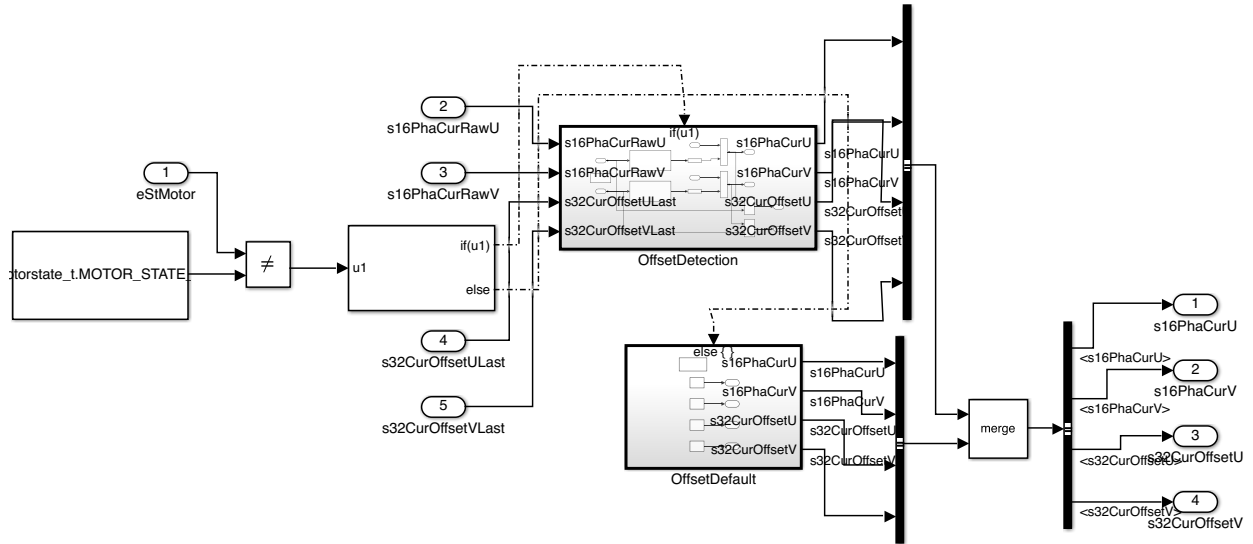
34. default

【default】 信号处理与功能传递模块。



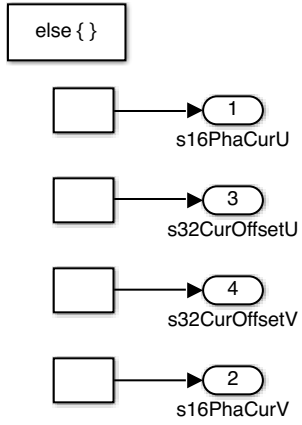
35. CurrentOffsets

【CurrentOffsets】信号处理与功能传递模块。



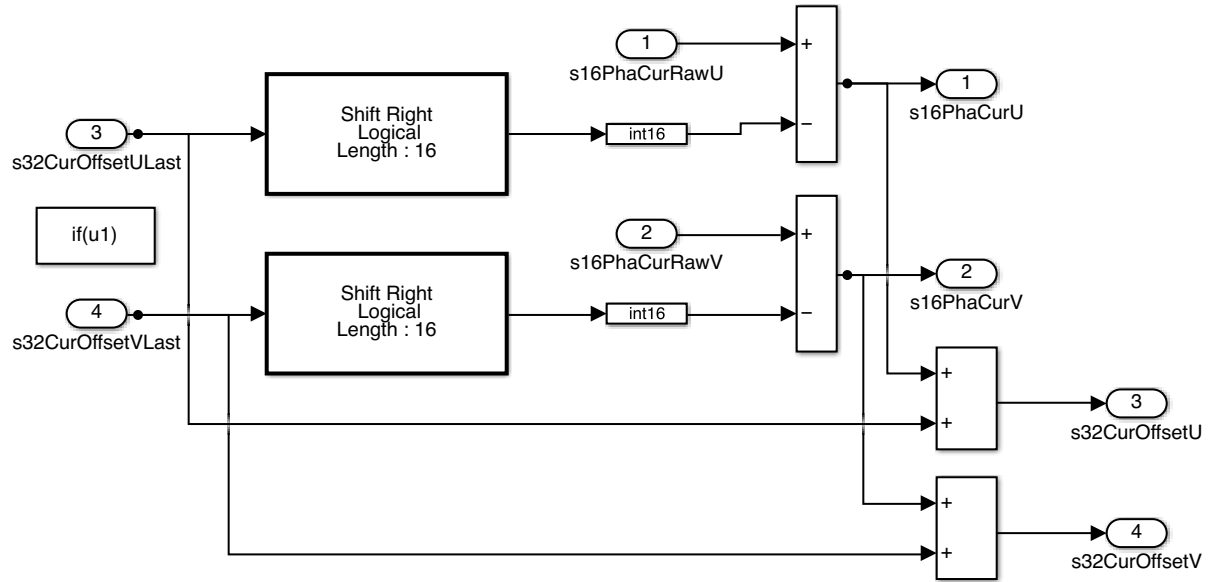
36. OffsetDefault

【OffsetDefault】 信号处理与功能传递模块。



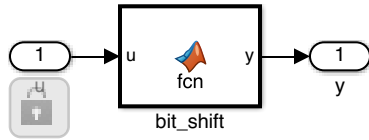
37. OffsetDetection

【OffsetDetection】 信号处理与功能传递模块。



38. BitShift

【BitShift】信号处理与功能传递模块。



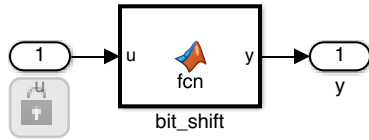
39. bit_shift

【bit_shift】 信号处理与功能传递模块。

Diagram snapshot unavailable: Specified diagram source "1.8841e+04" does not contain a diagram.

40. BitShift1

【BitShift1】信号处理与功能传递模块。

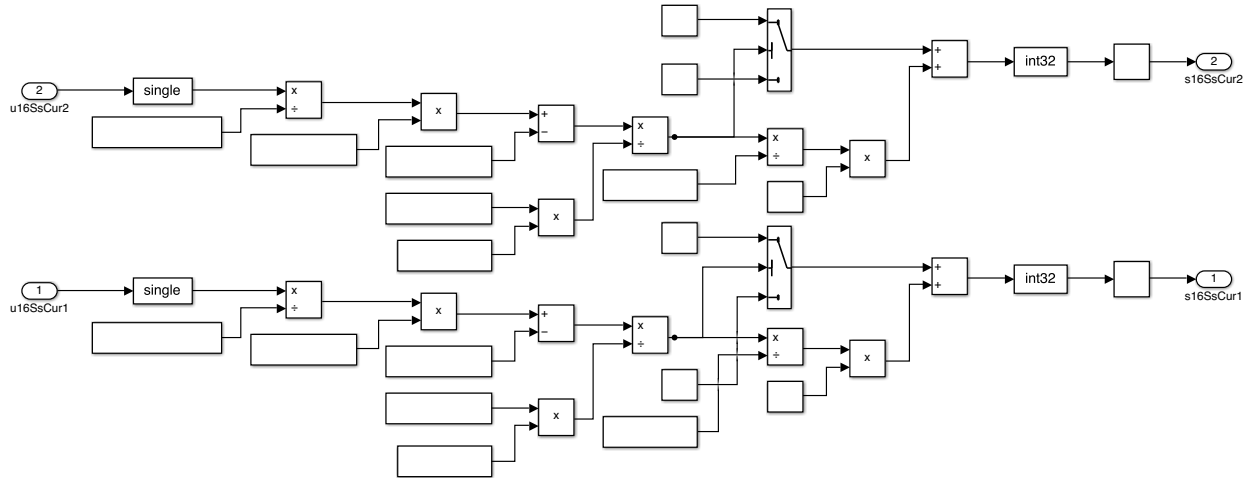


41. GetCurrents

【GetCurrents】信号处理与功能传递模块。

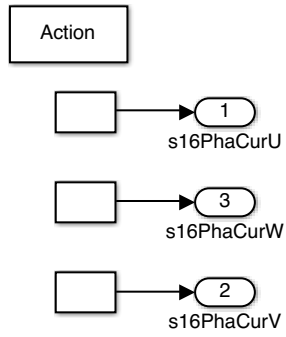
42. AdcVolToCur

【AdcVolToCur】信号处理与功能传递模块。



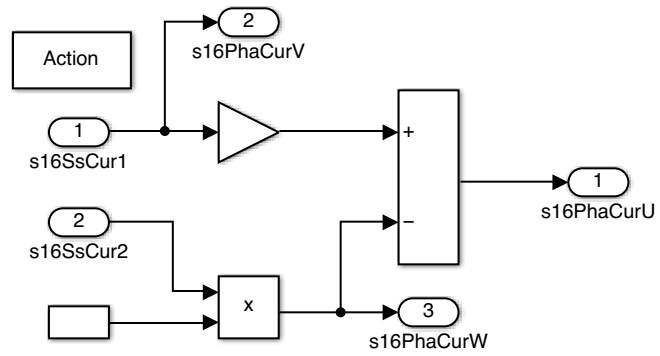
43. DEFAULT

【DEFAULT】 信号处理与功能传递模块。



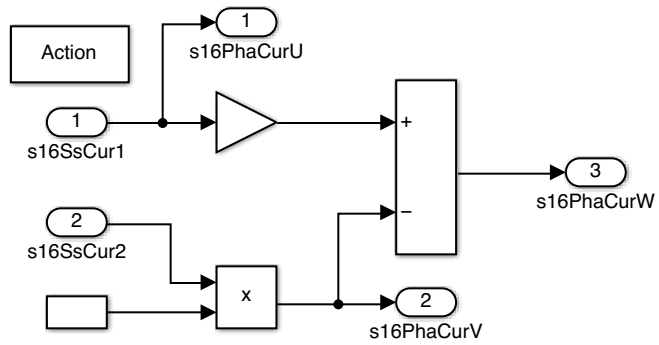
44. SECTOR_1

【SECTOR_1】 信号处理与功能传递模块。



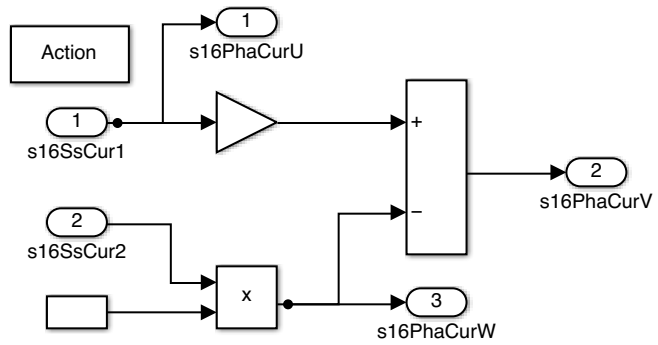
45. SECTOR_2

【SECTOR_2】信号处理与功能传递模块。



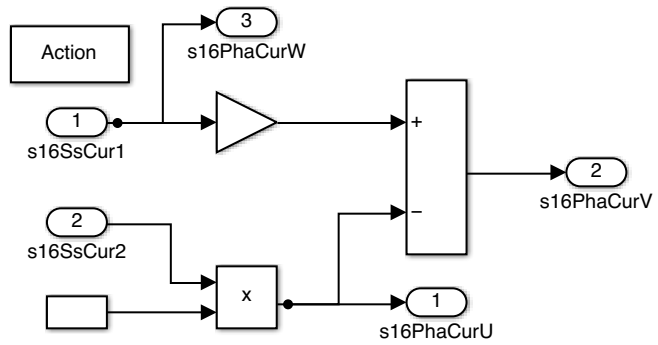
46. SECTOR_3

【SECTOR_3】信号处理与功能传递模块。



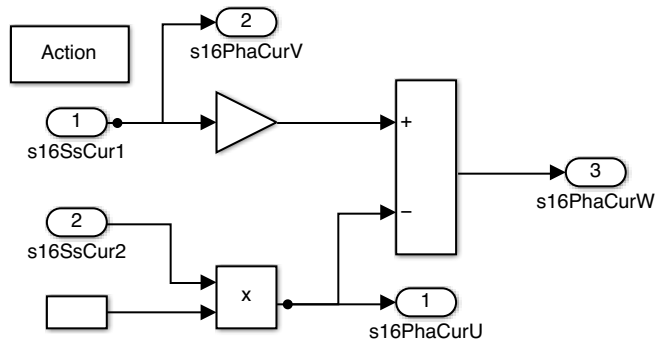
47. SECTOR_4

【SECTOR_4】信号处理与功能传递模块。



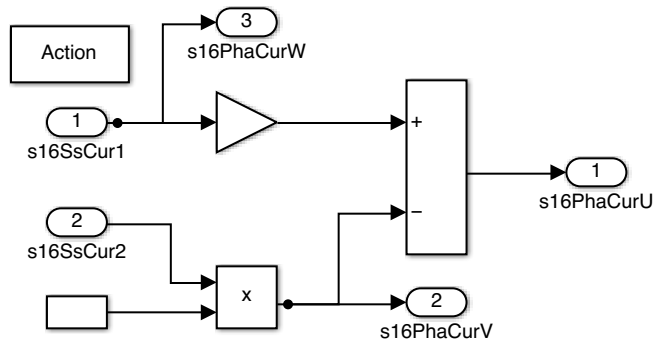
48. SECTOR_5

【SECTOR_5】信号处理与功能传递模块。



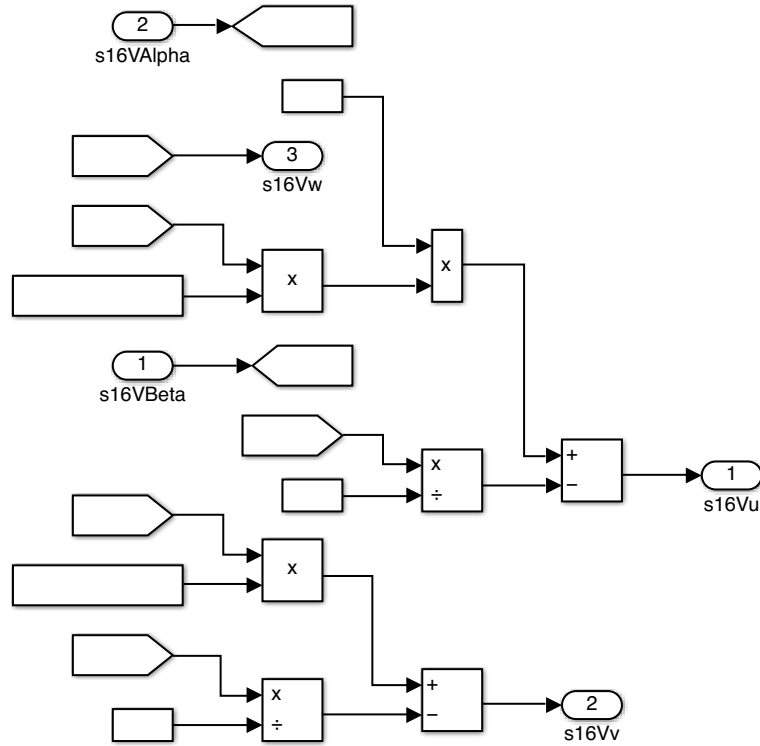
49. SECTOR_6

【SECTOR_6】信号处理与功能传递模块。



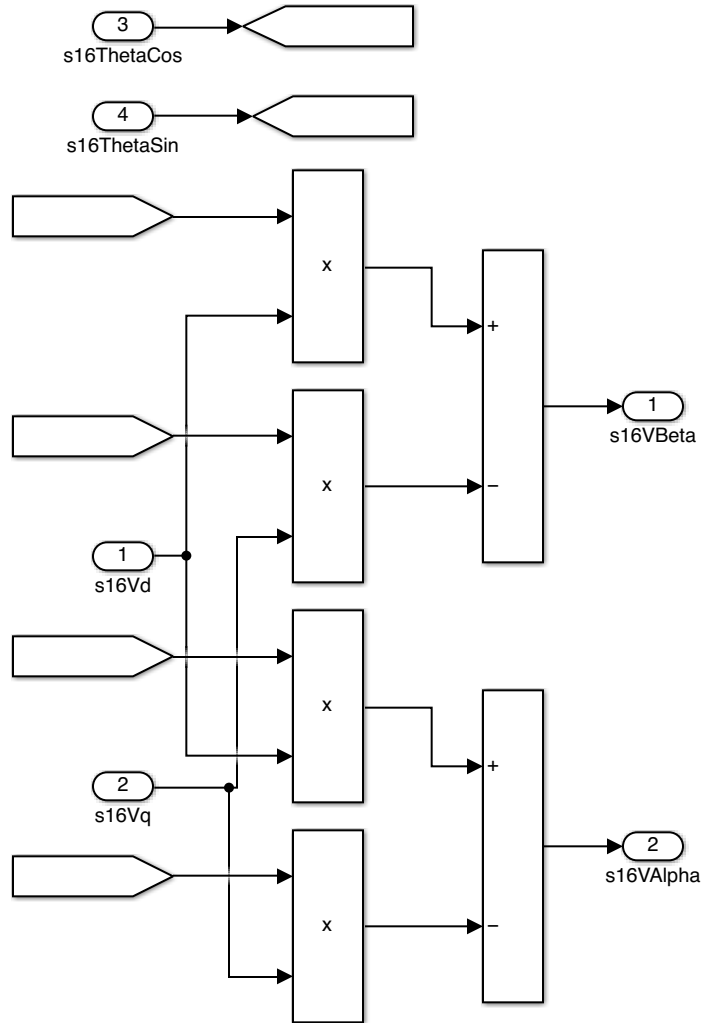
50. InverseClarkTransformations

【InverseClarkTransformations】信号处理与功能传递模块。



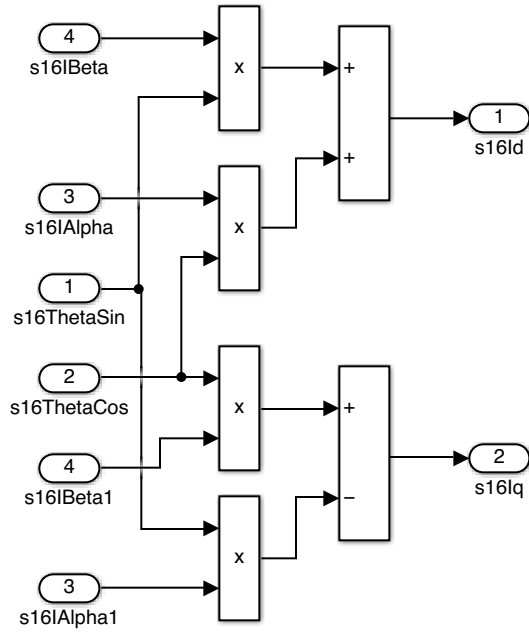
51. InverseParkTransformations

【InverseParkTransformations】信号处理与功能传递模块。



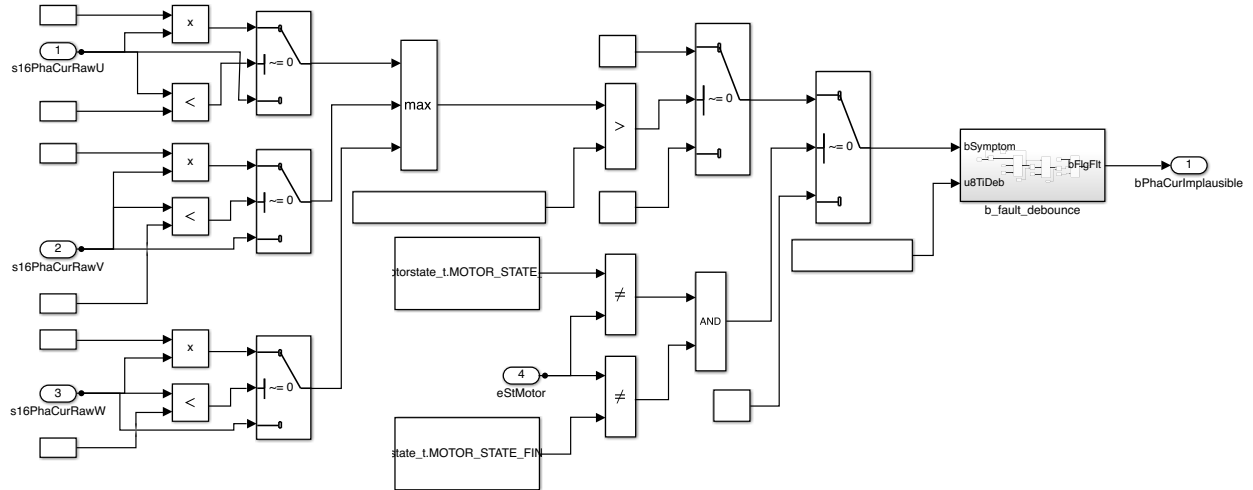
52. ParkTransformations

【ParkTransformations】信号处理与功能传递模块。



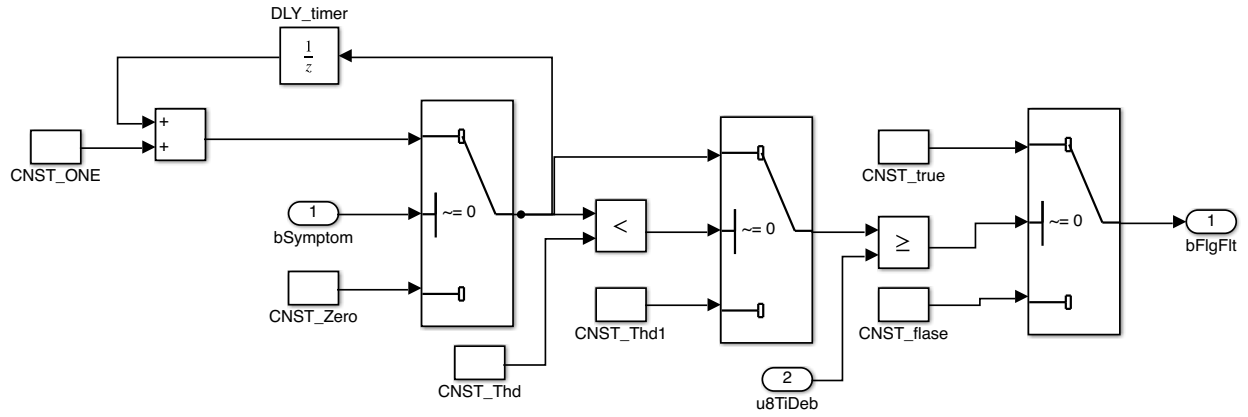
53. PhaseCurrentSpikeMonitor

【PhaseCurrentSpikeMonitor】信号处理与功能传递模块。



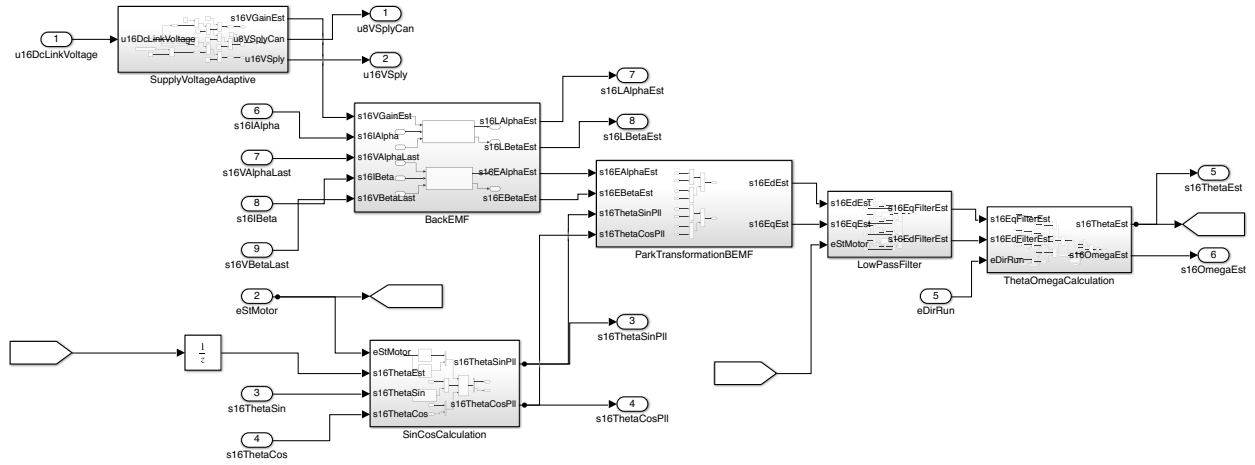
54. b_fault_debounce

【b_fault_debounce】信号处理与功能传递模块。



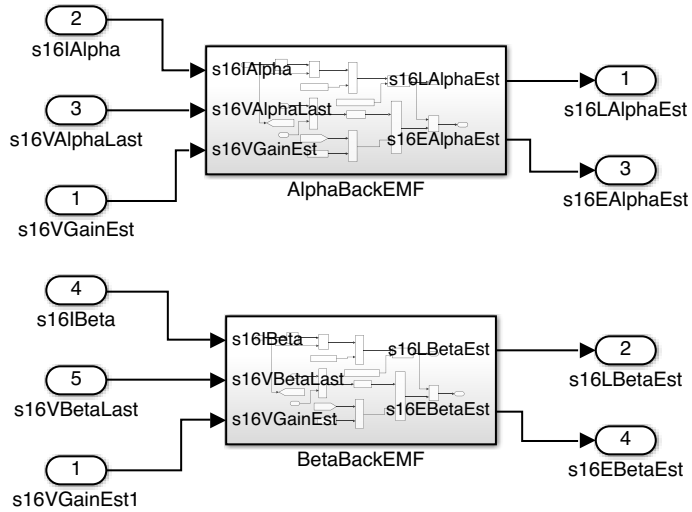
55. PIIObserver

【PIIObserver】信号处理与功能传递模块。



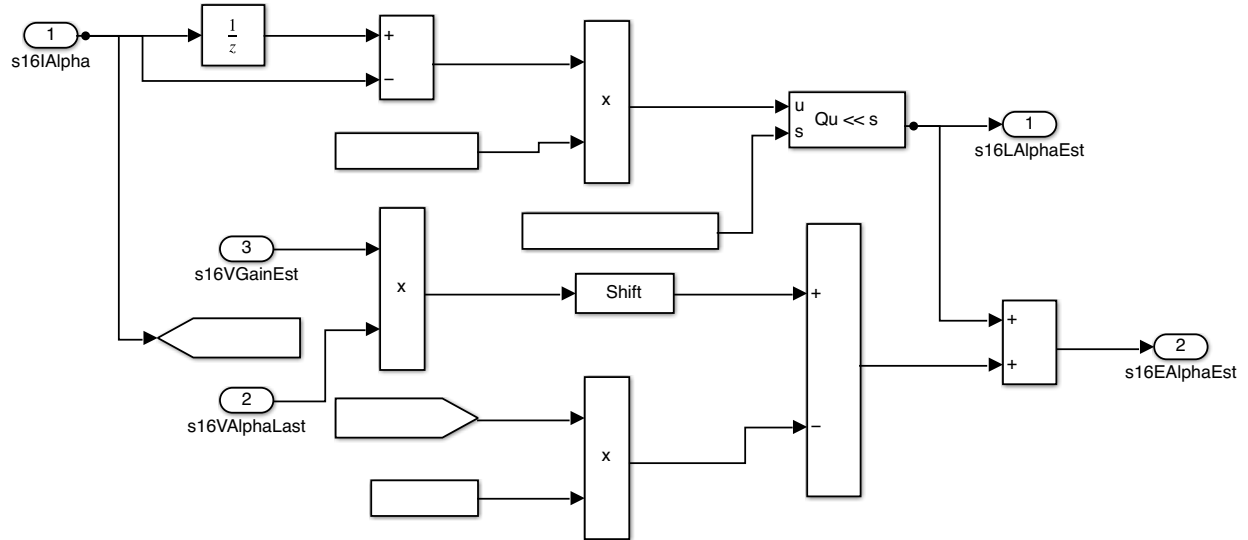
56. BackEMF

【BackEMF】信号处理与功能传递模块。



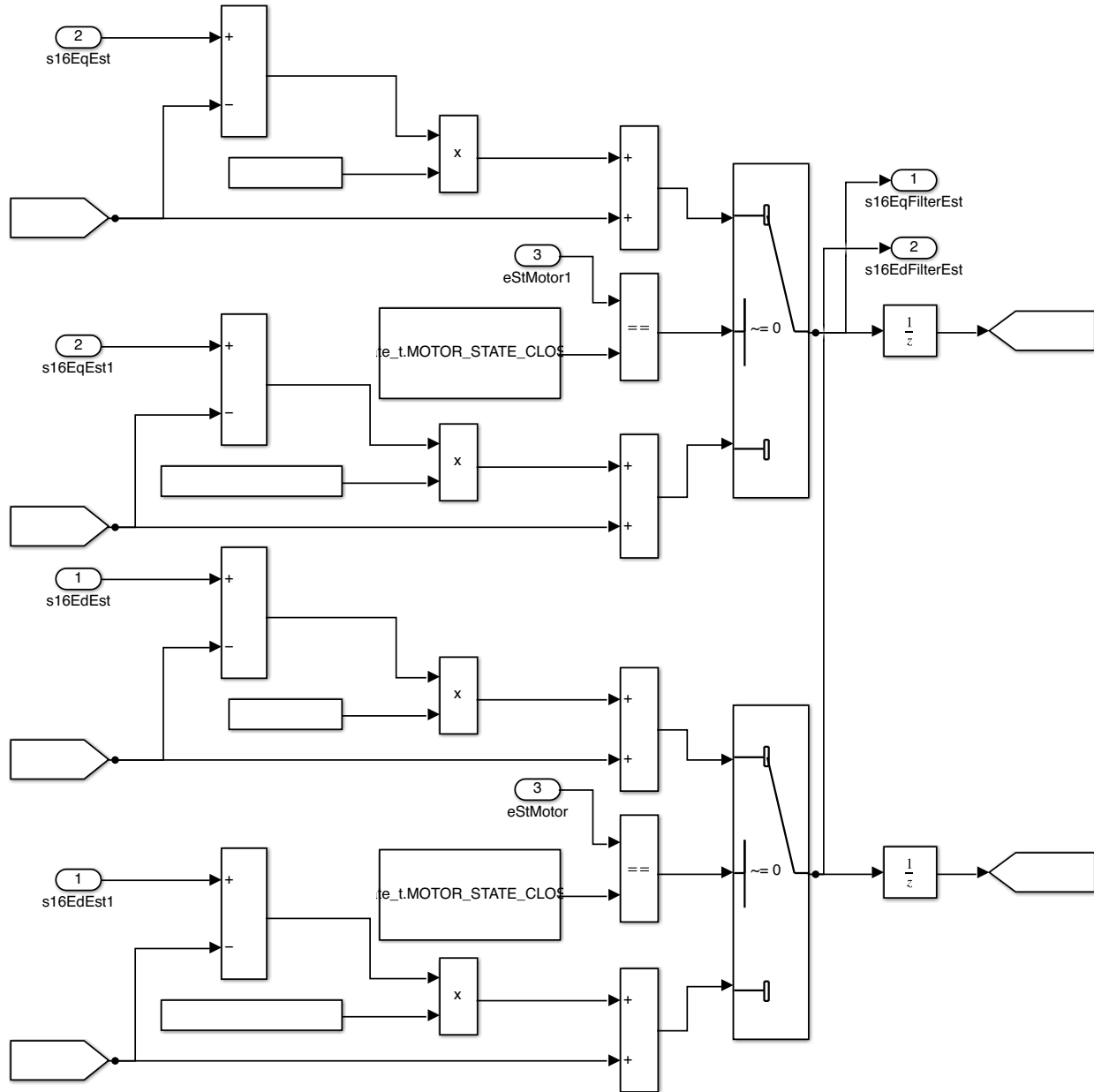
57. AlphaBackEMF

【AlphaBackEMF】信号处理与功能传递模块。



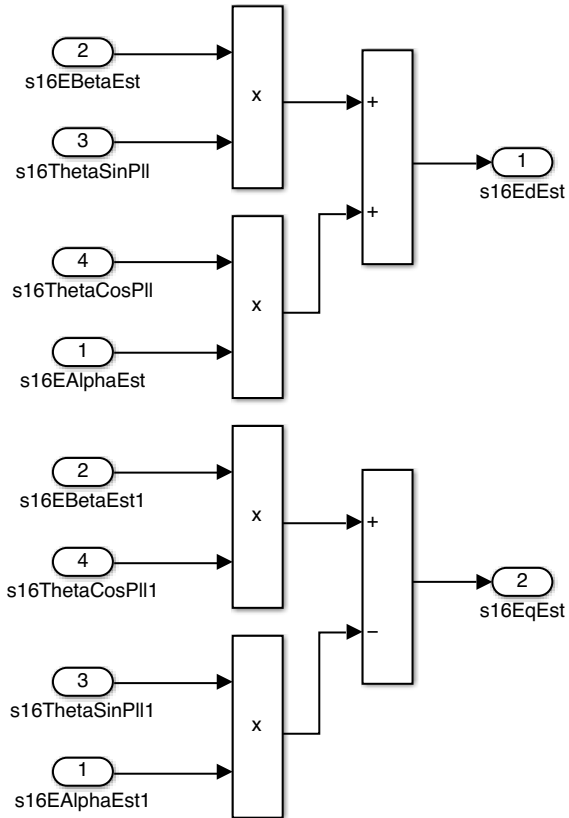
59. LowPassFilter

【LowPassFilter】信号处理与功能传递模块。



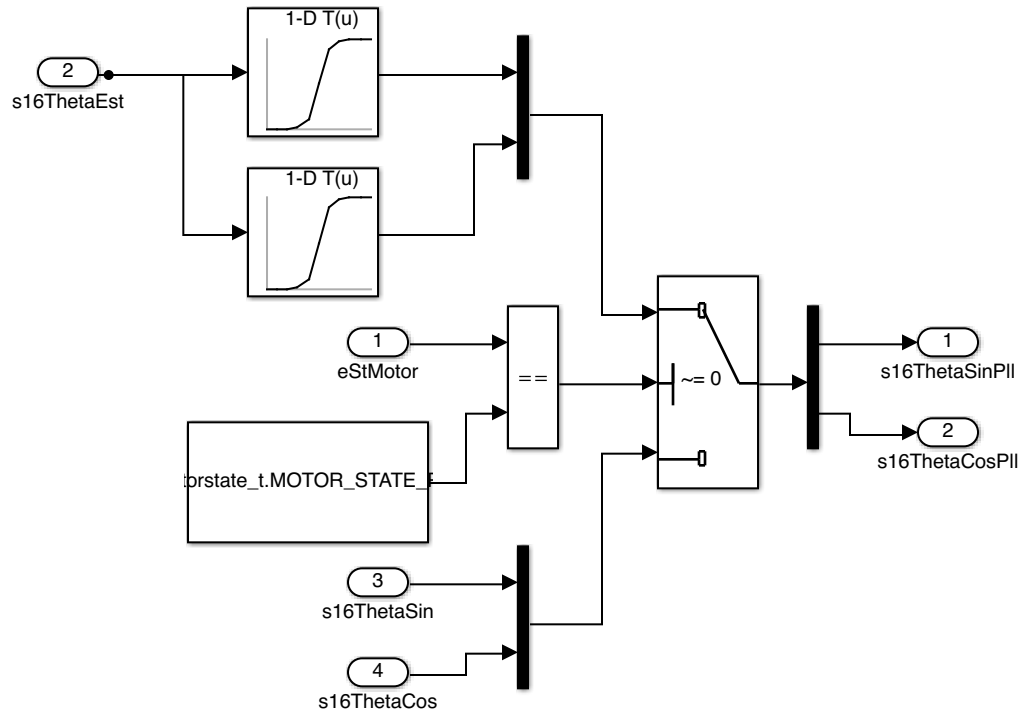
60. ParkTransformationBEMF

【ParkTransformationBEMF】信号处理与功能传递模块。



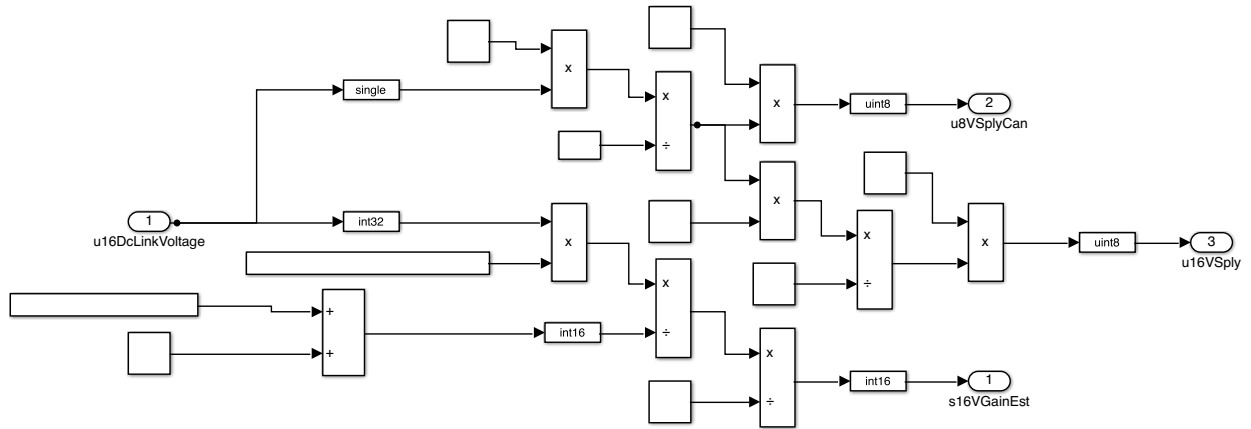
61. SinCosCalculation

【SinCosCalculation】 信号处理与功能传递模块。



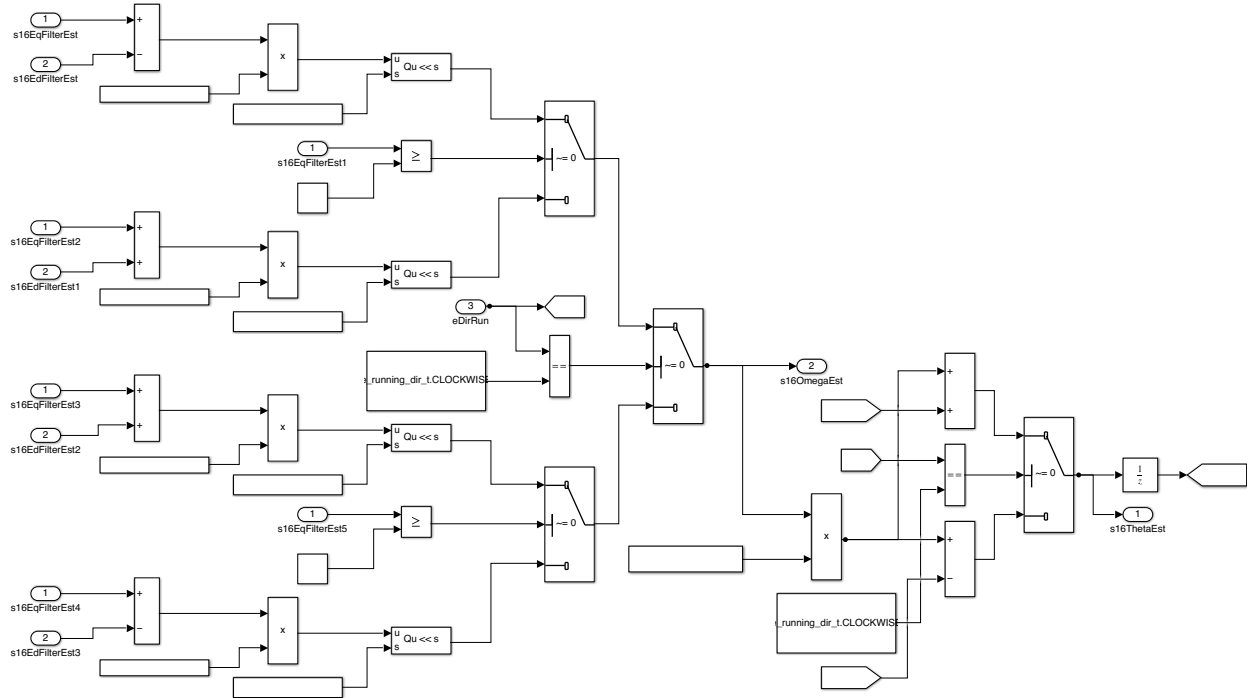
62. SupplyVoltageAdaptive

【SupplyVoltageAdaptive】信号处理与功能传递模块。



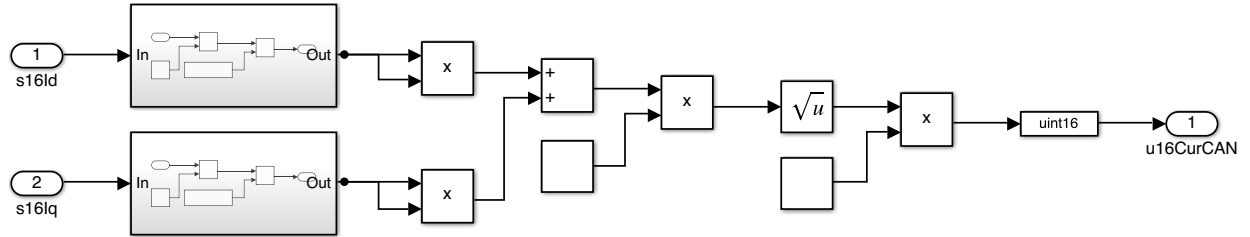
63. ThetaOmegaCalculation

【ThetaOmegaCalculation】信号处理与功能传递模块。



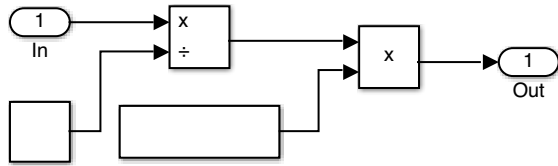
64. SendCurrentToCan

【SendCurrentToCan】信号处理与功能传递模块。



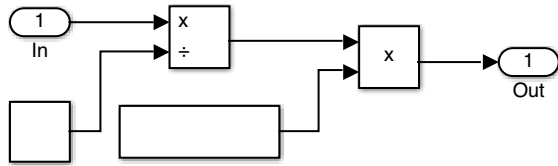
65. Subsystem

【Subsystem】 信号处理与功能传递模块。



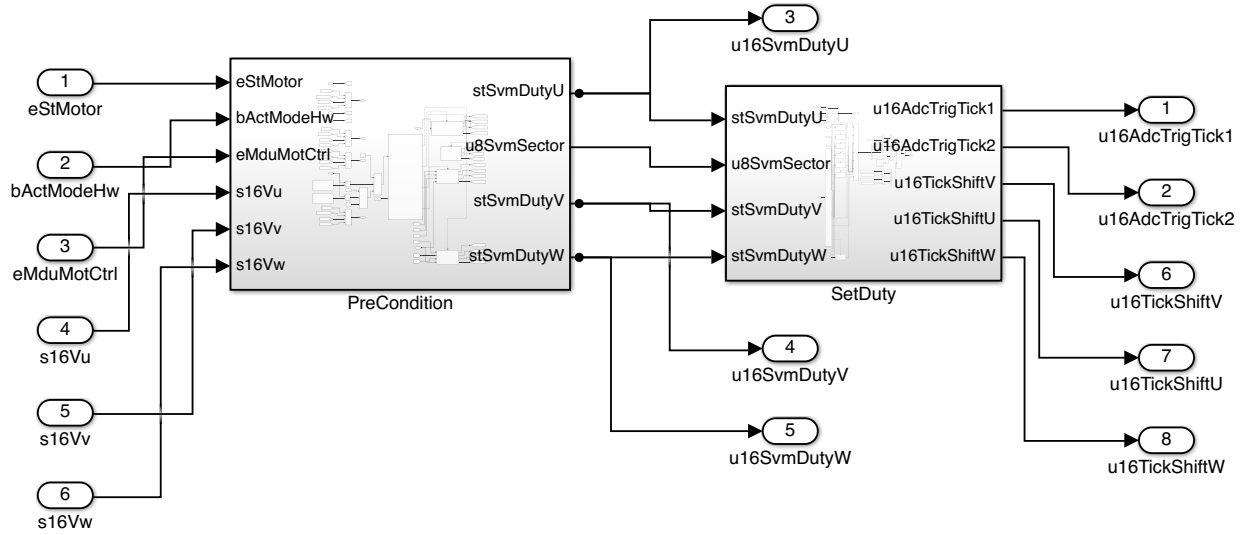
66. Subsystem1

【Subsystem1】 信号处理与功能传递模块。



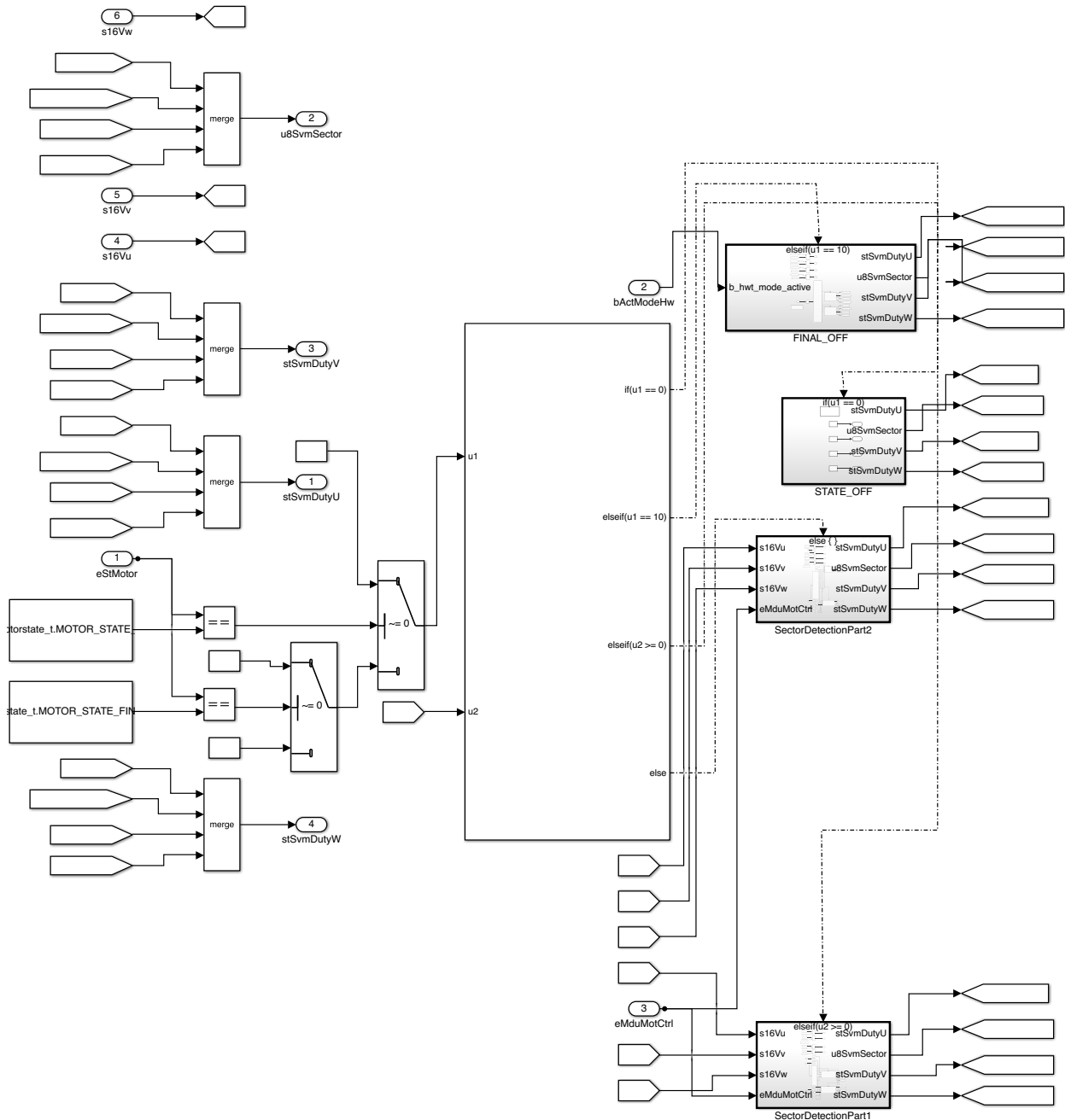
67. SetDutyCycle

【SetDutyCycle】信号处理与功能传递模块。



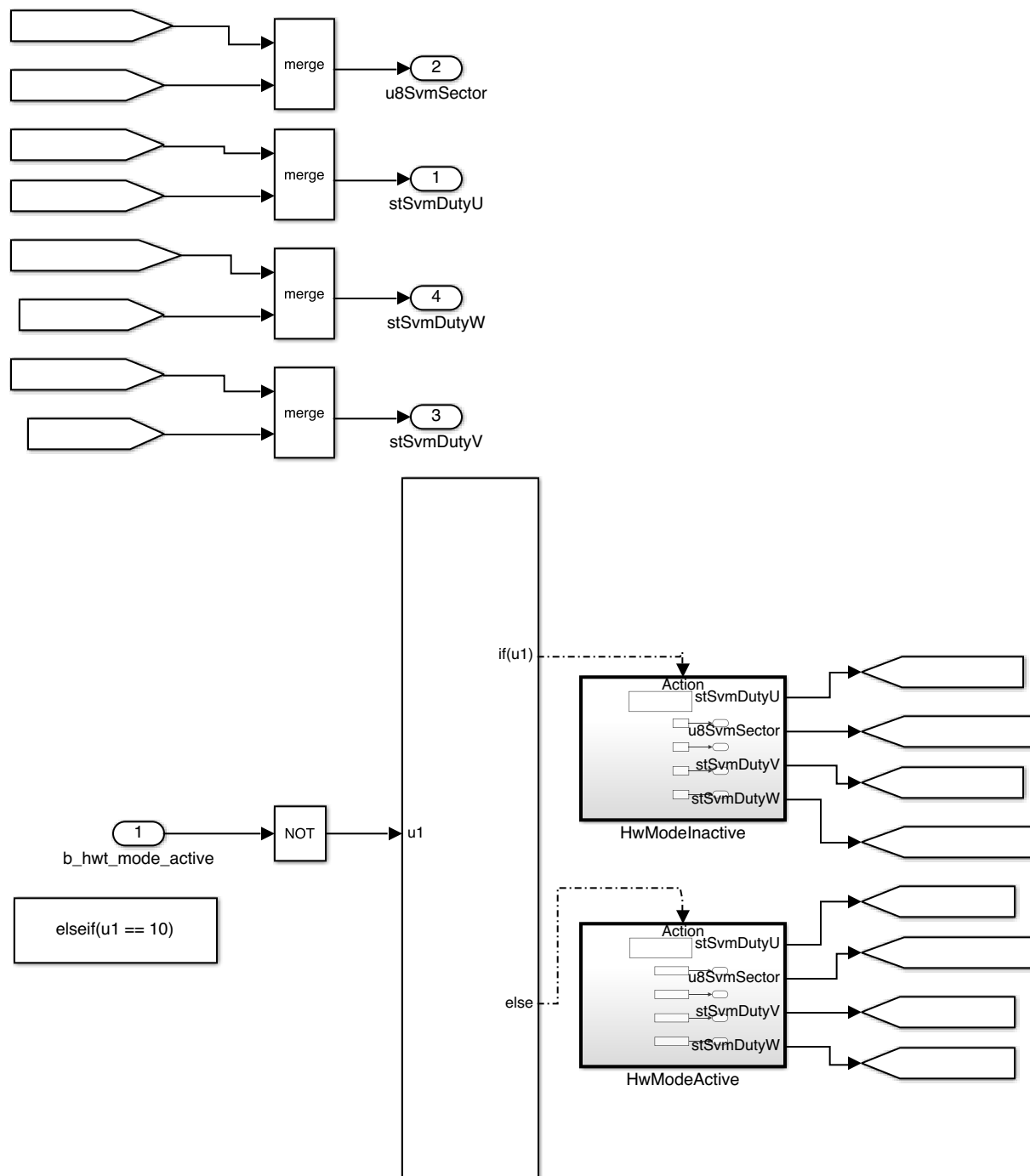
68. PreCondition

【PreCondition】 信号处理与功能传递模块。



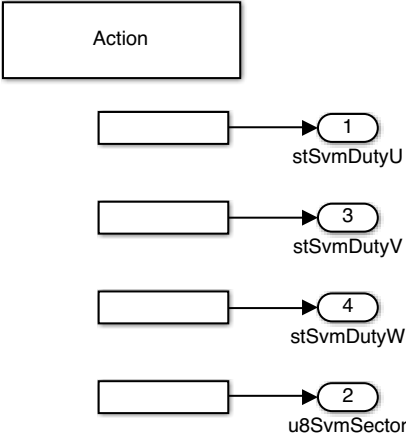
69. FINAL_OFF

【FINAL_OFF】信号处理与功能传递模块。



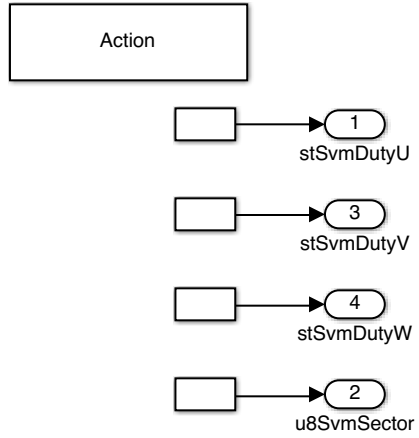
70. HwModeActive

【HwModeActive】 信号处理与功能传递模块。



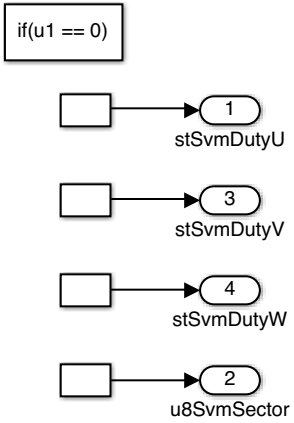
71. HwModeInactive

【HwModeInactive】信号处理与功能传递模块。



72. STATE_OFF

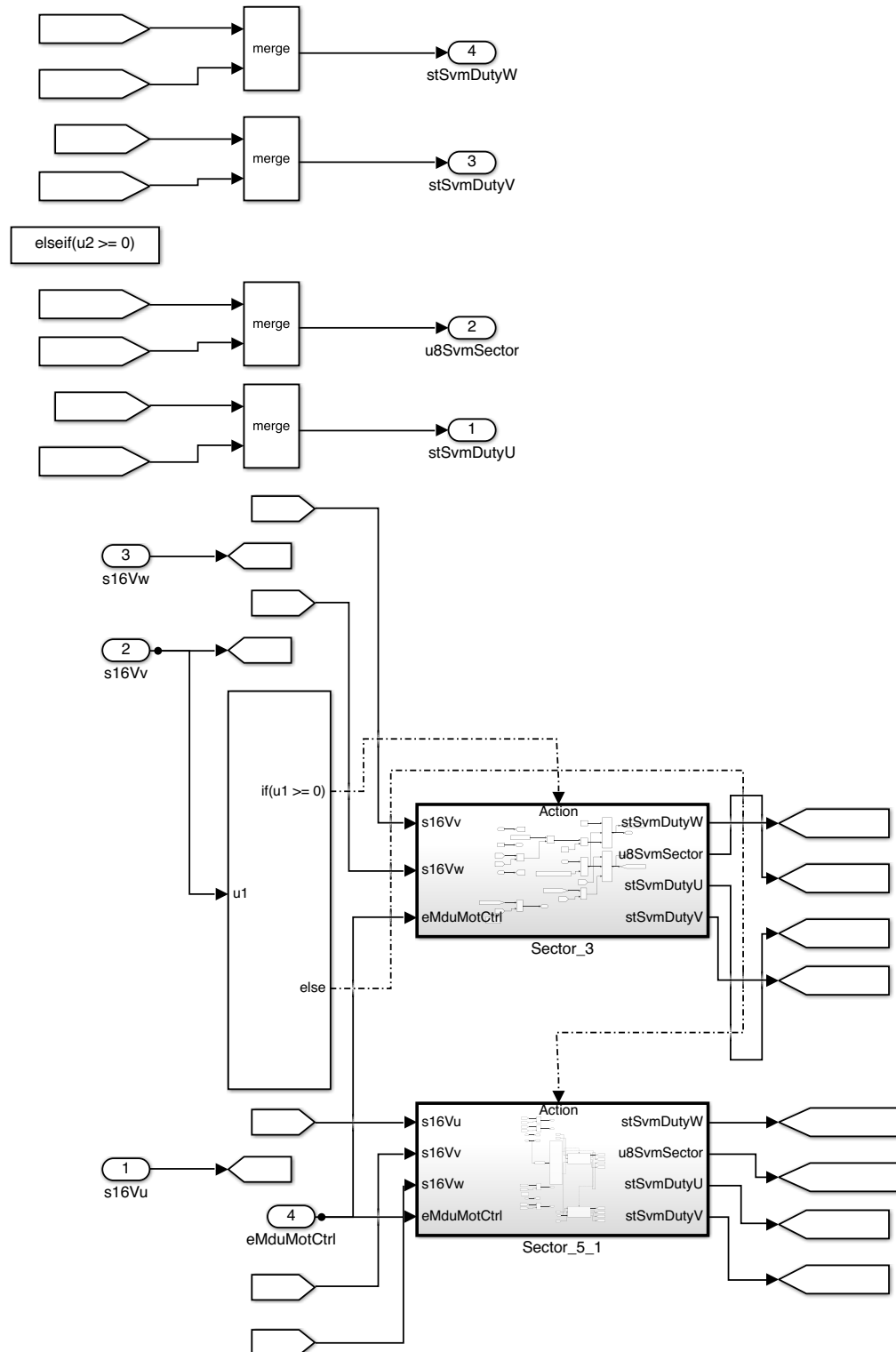
【STATE_OFF】 信号处理与功能传递模块。



73. SectorDetectionPart1

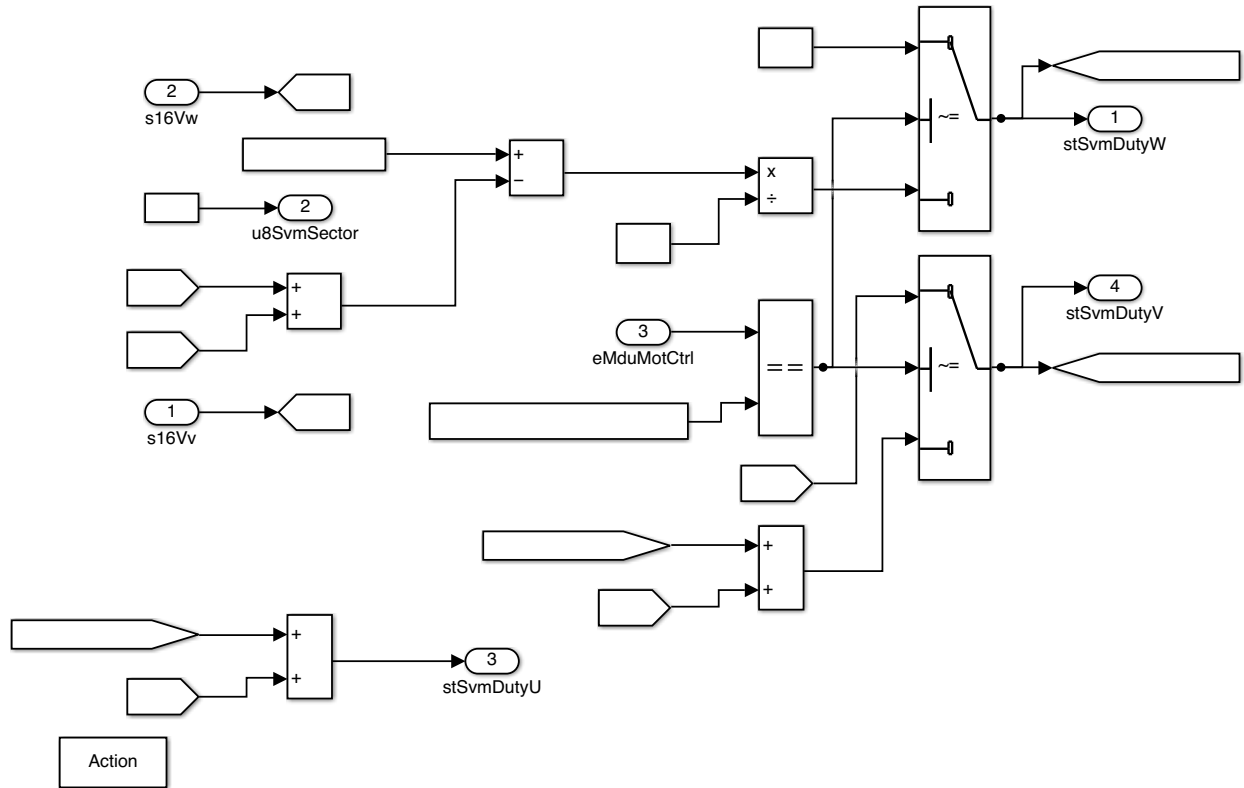
【SectorDetectionPart1】 信号处理与功能传递模块。

73. SectorDetectionPart1



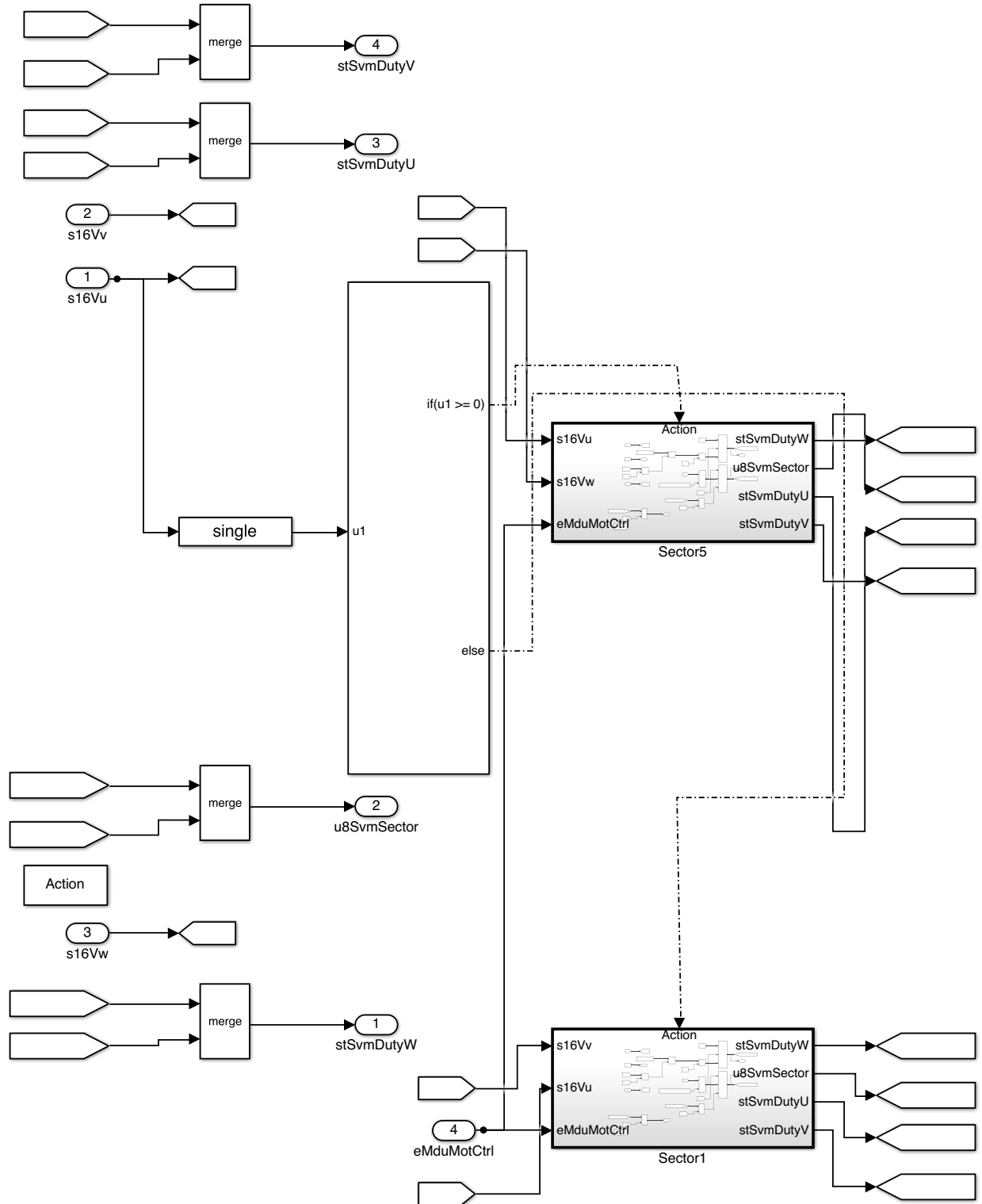
74. Sector_3

【Sector_3】 信号处理与功能传递模块。



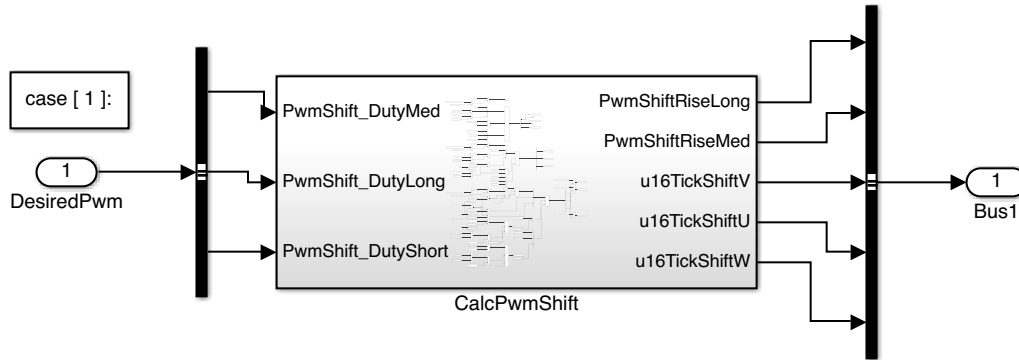
75. Sector_5_1

【Sector_5_1】信号处理与功能传递模块。



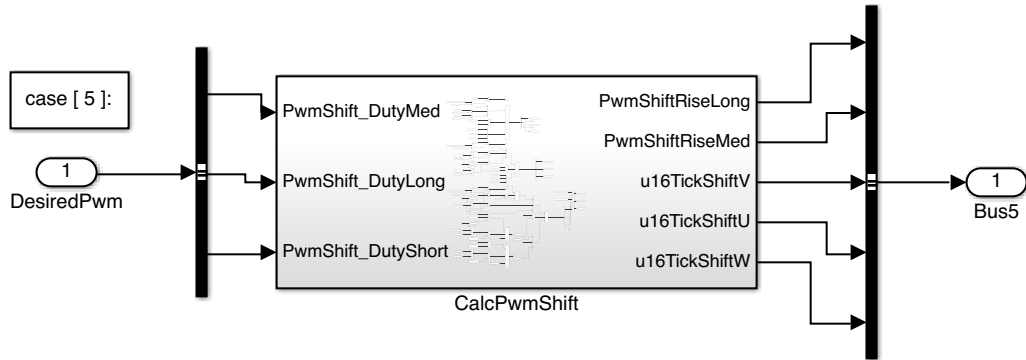
76. Sector1

【Sector1】 信号处理与功能传递模块。



77. Sector5

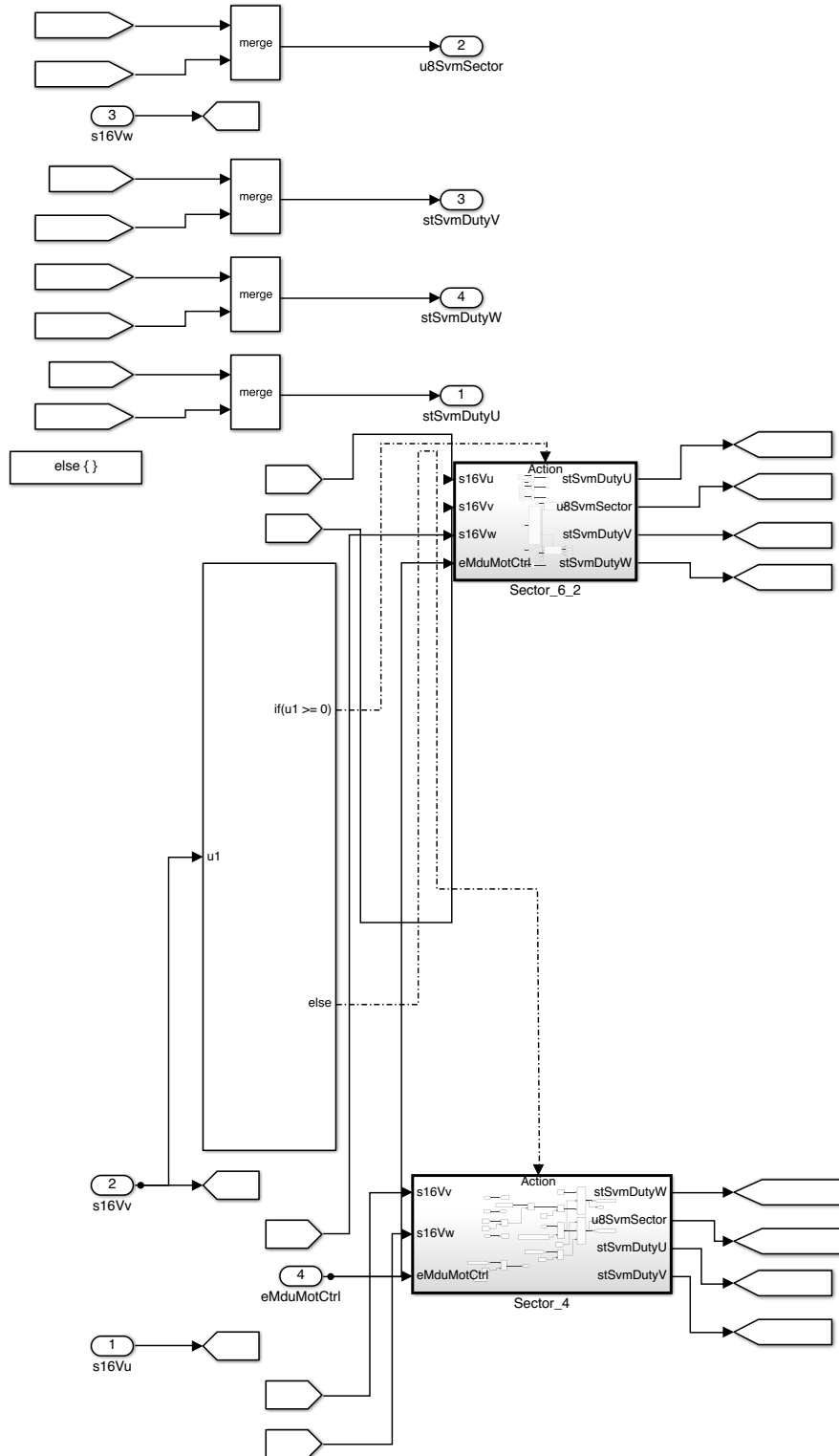
【Sector5】信号处理与功能传递模块。



78. SectorDetectionPart2

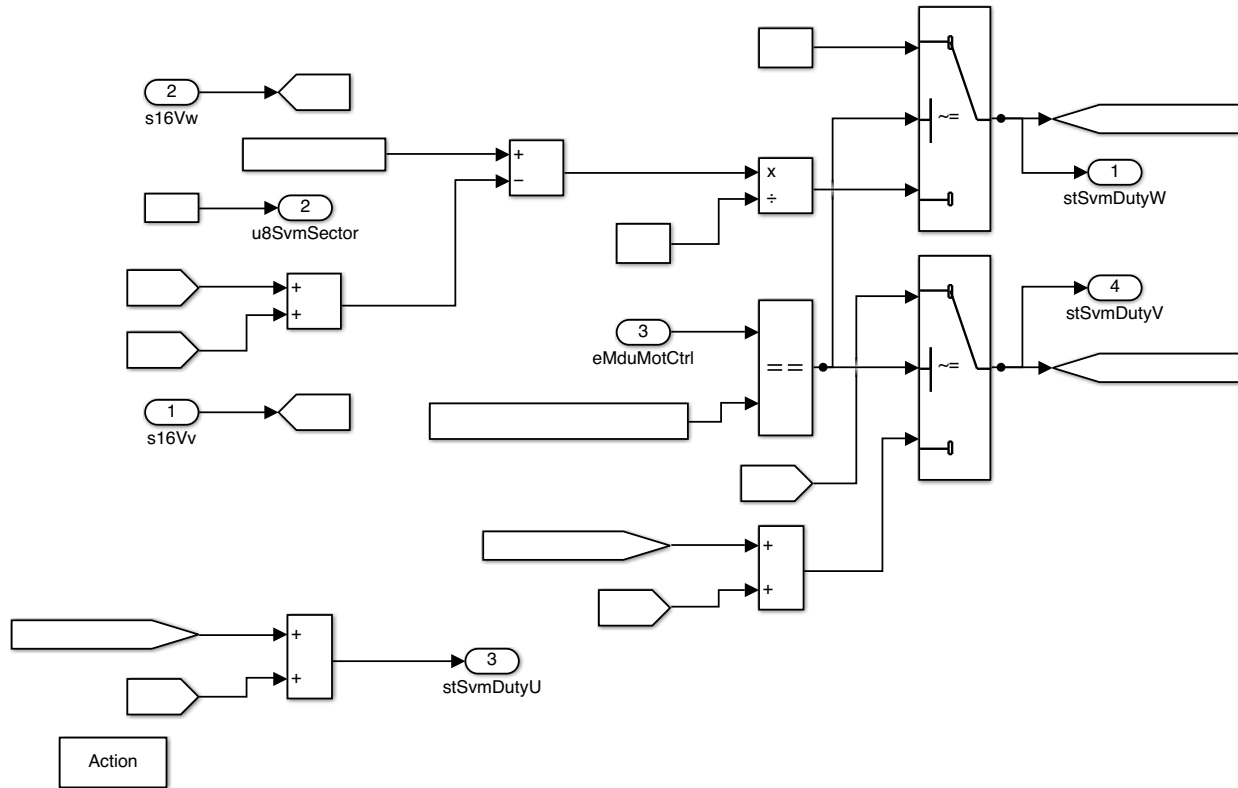
【SectorDetectionPart2】信号处理与功能传递模块。

78. SectorDetectionPart2



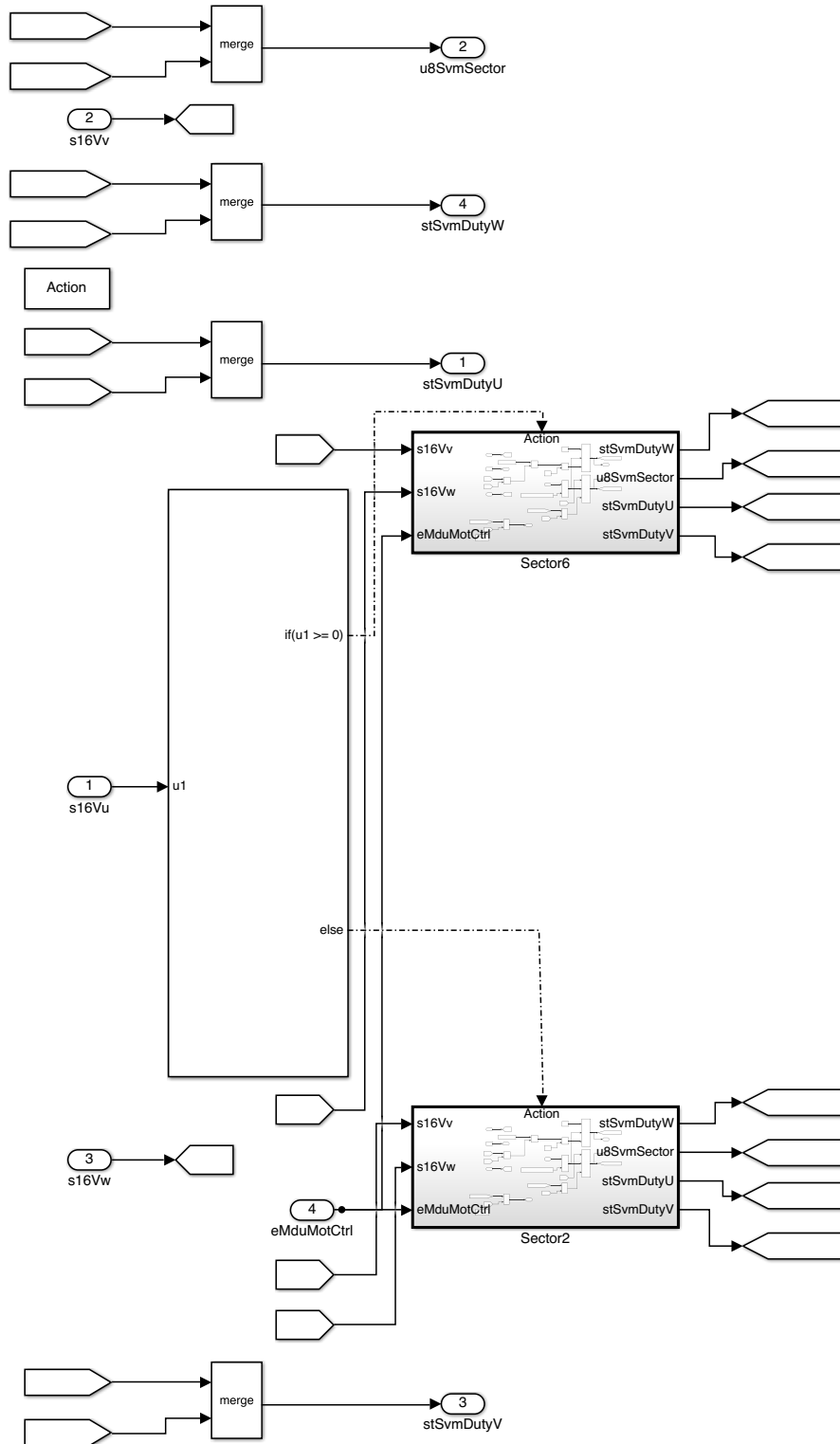
79. Sector_4

【Sector_4】 信号处理与功能传递模块。



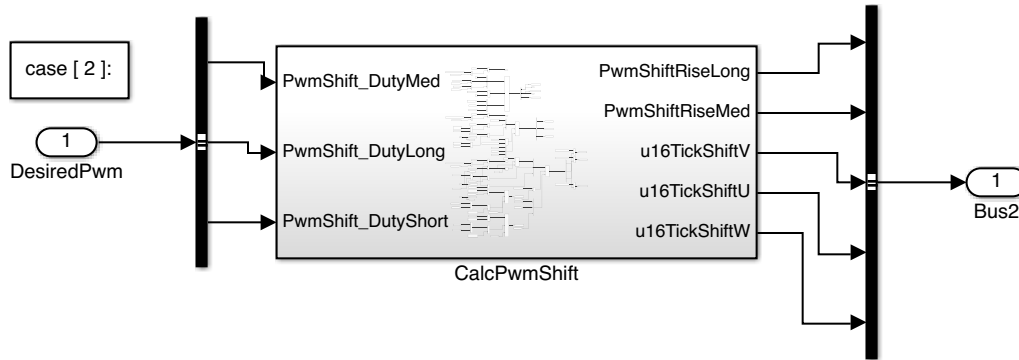
80. Sector_6_2

【Sector_6_2】信号处理与功能传递模块。



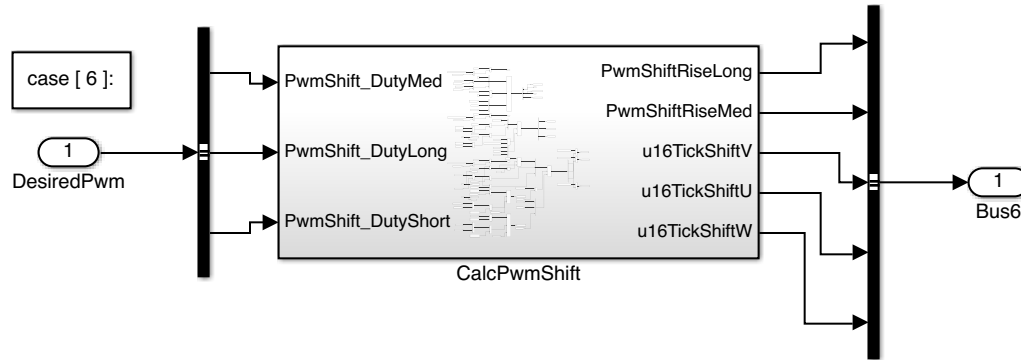
81. Sector2

【Sector2】 信号处理与功能传递模块。



82. Sector6

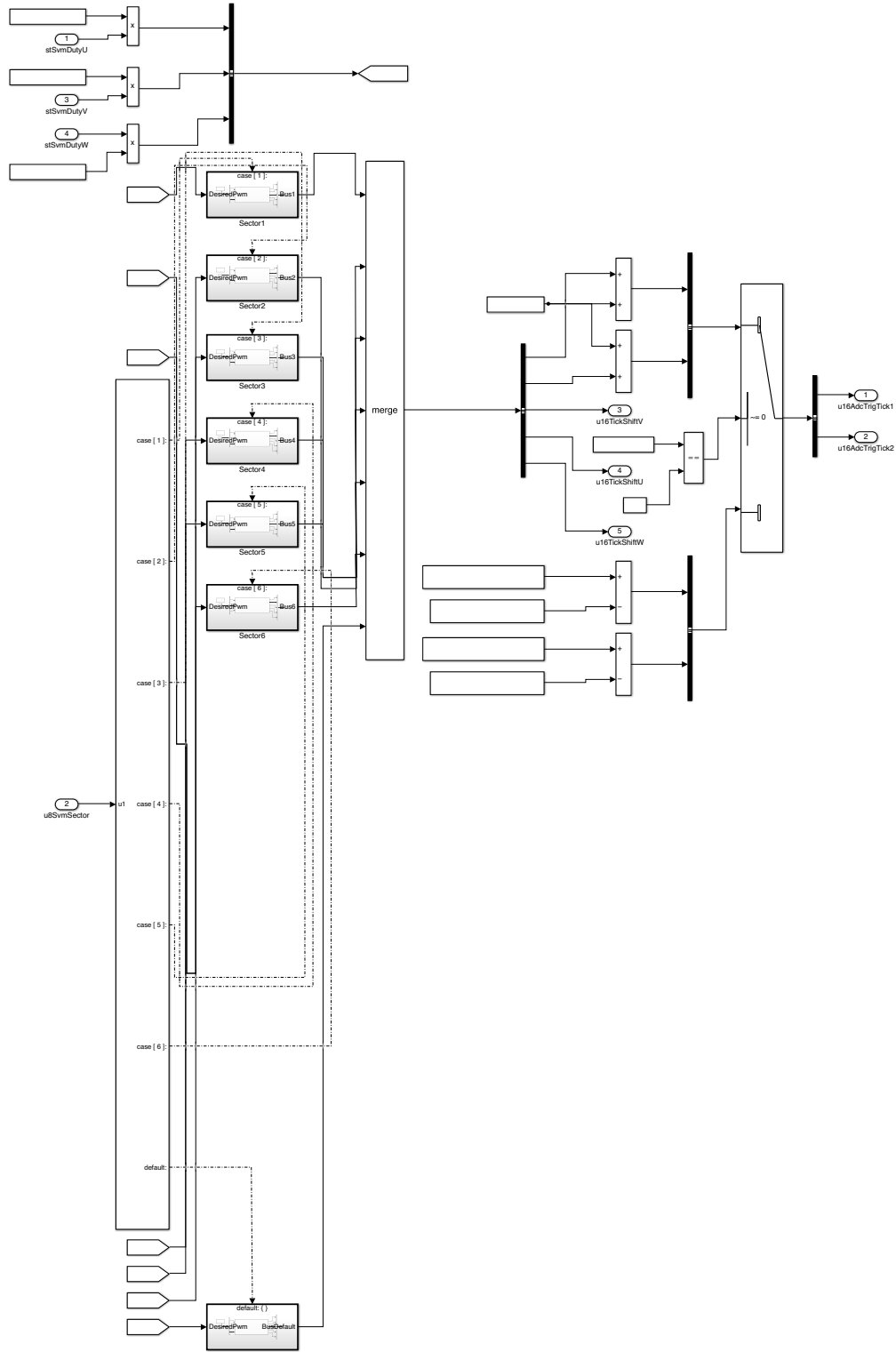
【Sector6】 信号处理与功能传递模块。



83. SetDuty

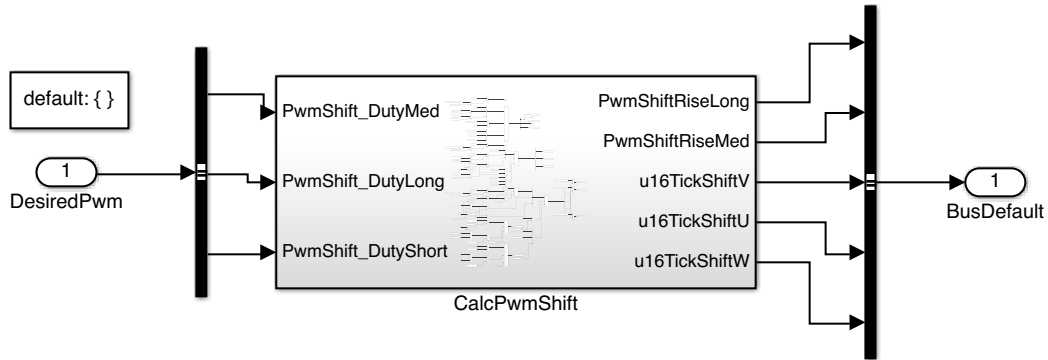
【SetDuty】信号处理与功能传递模块。

83. SetDuty



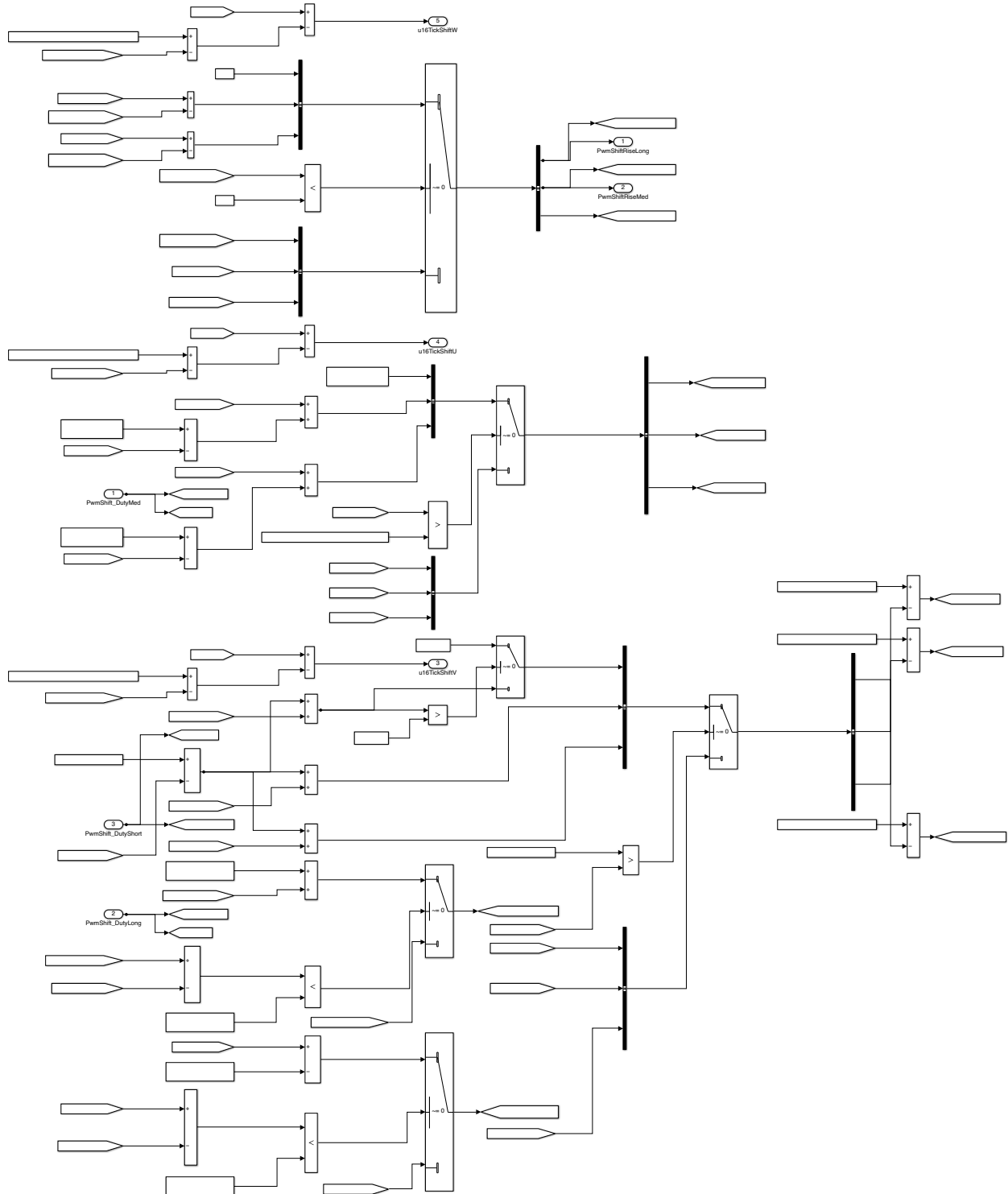
84. Default

【Default】信号处理与功能传递模块。



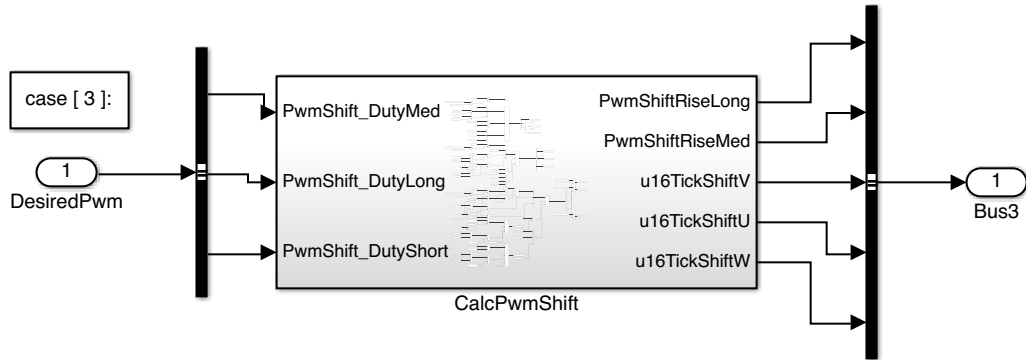
85. CalcPwmShift

【CalcPwmShift】信号处理与功能传递模块。



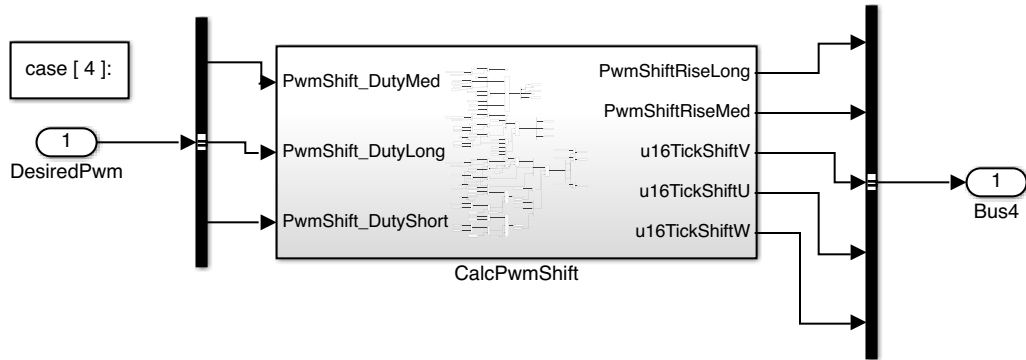
86. Sector3

【Sector3】 信号处理与功能传递模块。



87. Sector4

【Sector4】 信号处理与功能传递模块。



88. foc_calc_sin_table

【foc_calc_sin_table】信号处理与功能传递模块。

